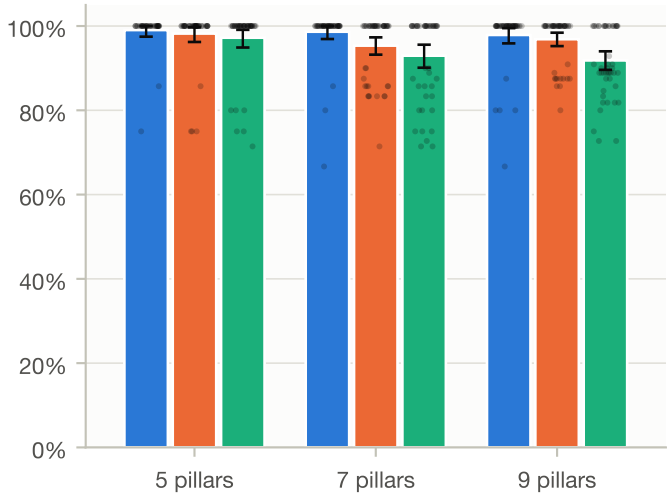
Cloth-drop experiment: does the cloth find MWT edges?

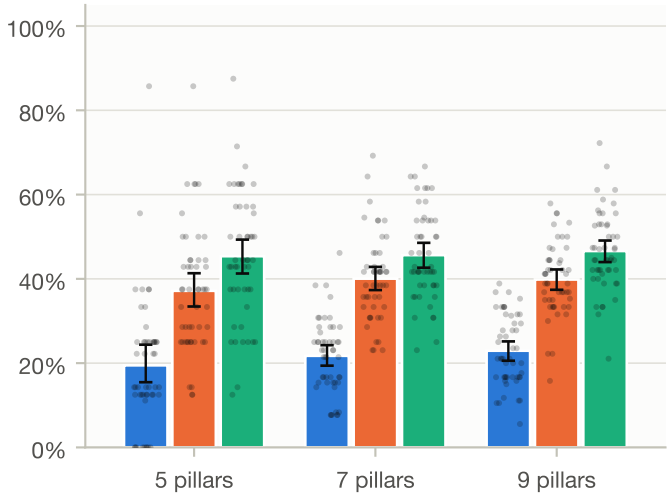
450 trials: 3 cloth types x 3 pillar counts x 50 random layouts (the same layouts for every cloth type). Bars: mean with 95% bootstrap interval; dots: single trials.

Precision: cloth edges that are MWT edges



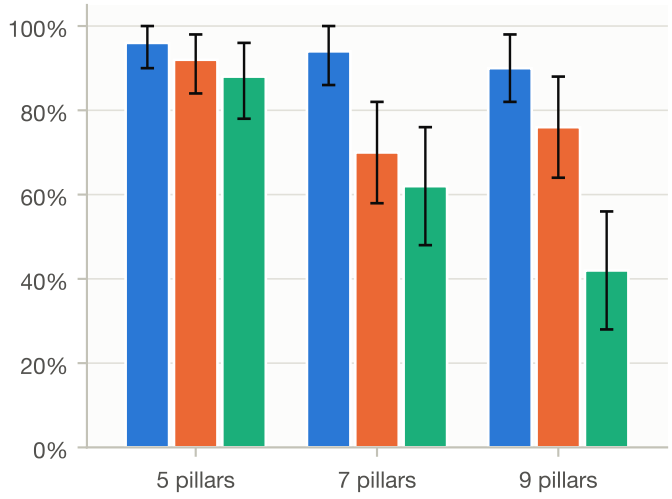
100% = every ridge the cloth formed is an edge of the minimum weight triangulation.

Recall: MWT edges the cloth found



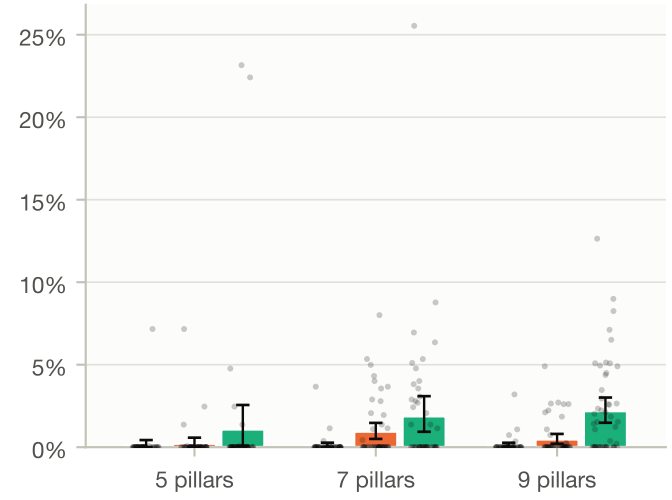
How much of the MWT the cloth builds by itself; the rest must be completed.

Trials where the cloth path is inside an MWT



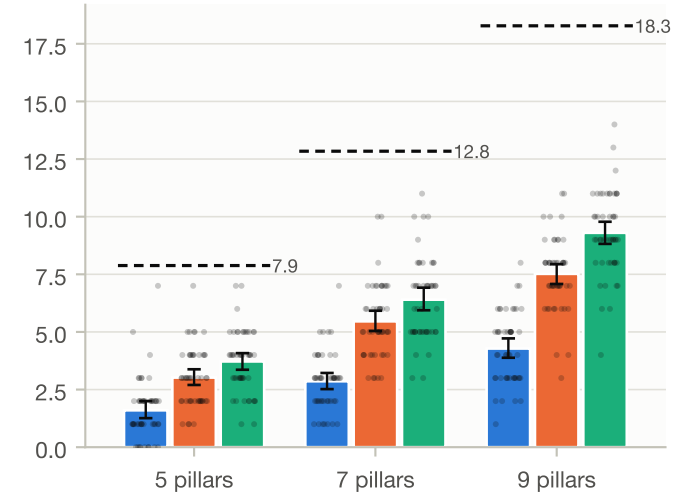
The cloth made no wrong choice: completing its edges gives an exact MWT.

Completion gap above the MWT weight



Lightest triangulation that keeps every cloth edge, compared with the MWT. 0% = optimal.

Edges formed by the cloth



Dashed: edges in a full triangulation (3n - 3 - hull points) for comparison.

Means per case

cloth, pillars

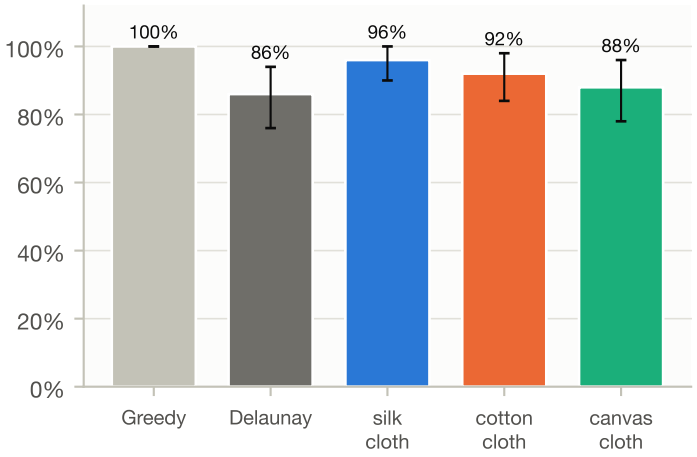
	Precision	Recall	In an MWT	Gap
silk, 5	99%	19%	96%	0.14%
cotton, 5	98%	37%	92%	0.22%
canvas, 5	97%	45%	88%	1.08%
silk, 7	99%	22%	94%	0.10%
cotton, 7	95%	40%	70%	0.95%
canvas, 7	93%	46%	62%	1.88%
silk, 9	98%	23%	90%	0.11%
cotton, 9	97%	40%	76%	0.48%
canvas, 9	92%	47%	42%	2.19%

Cloth vs. Greedy vs. Delaunay: which one gives the MWT?

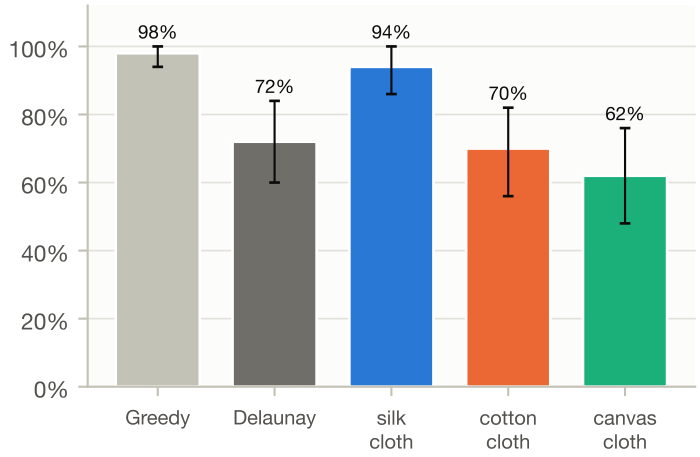
Cloth: the cloth's edges completed by Unity's exact solver (its yellow edges). Greedy and Delaunay are computed on the same pillar positions, with no search. Same layouts for every method.

silk cotton canvas Delaunay Greedy

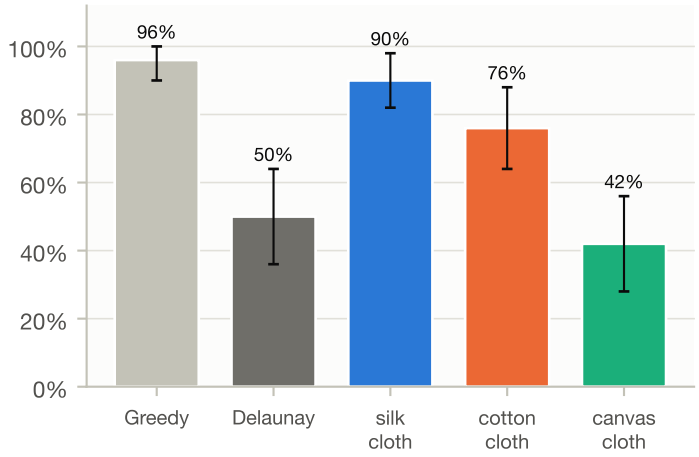
Trials that ARE the MWT, 5 pillars



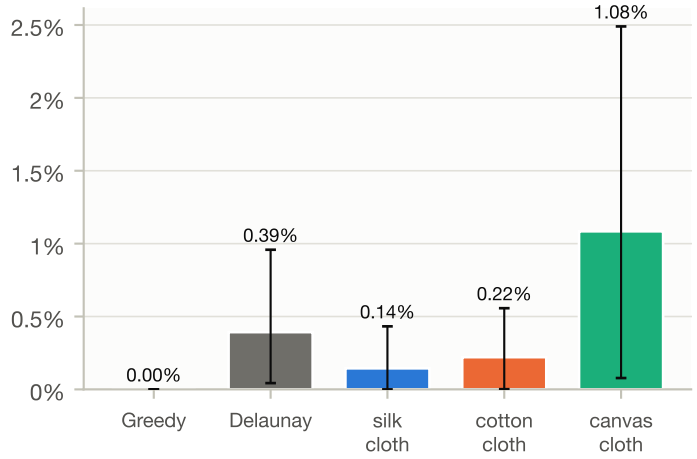
Trials that ARE the MWT, 7 pillars



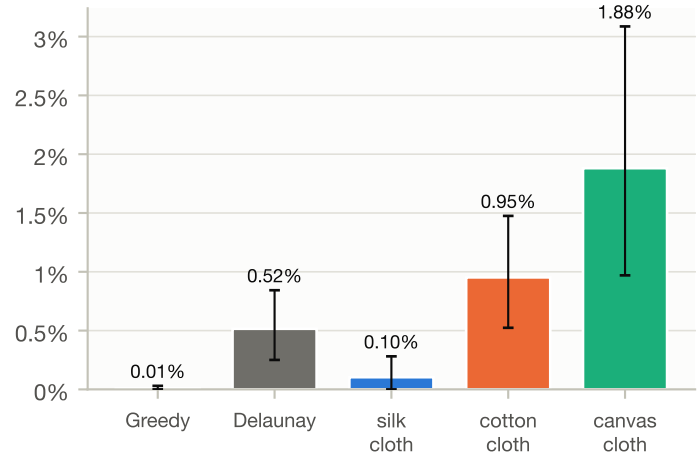
Trials that ARE the MWT, 9 pillars



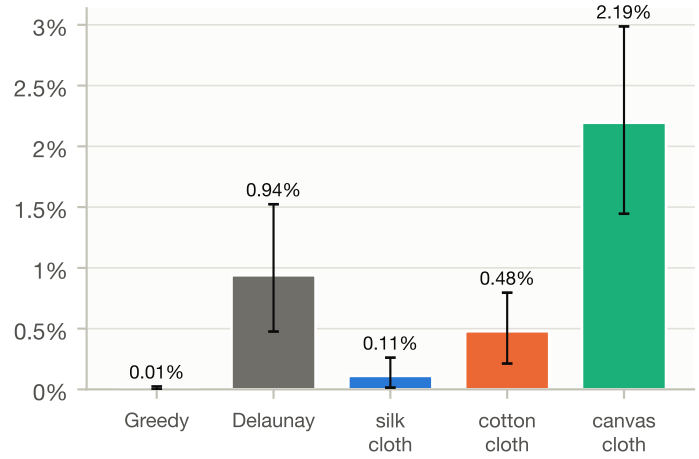
Mean gap above the MWT, 5 pillars



Mean gap above the MWT, 7 pillars



Mean gap above the MWT, 9 pillars



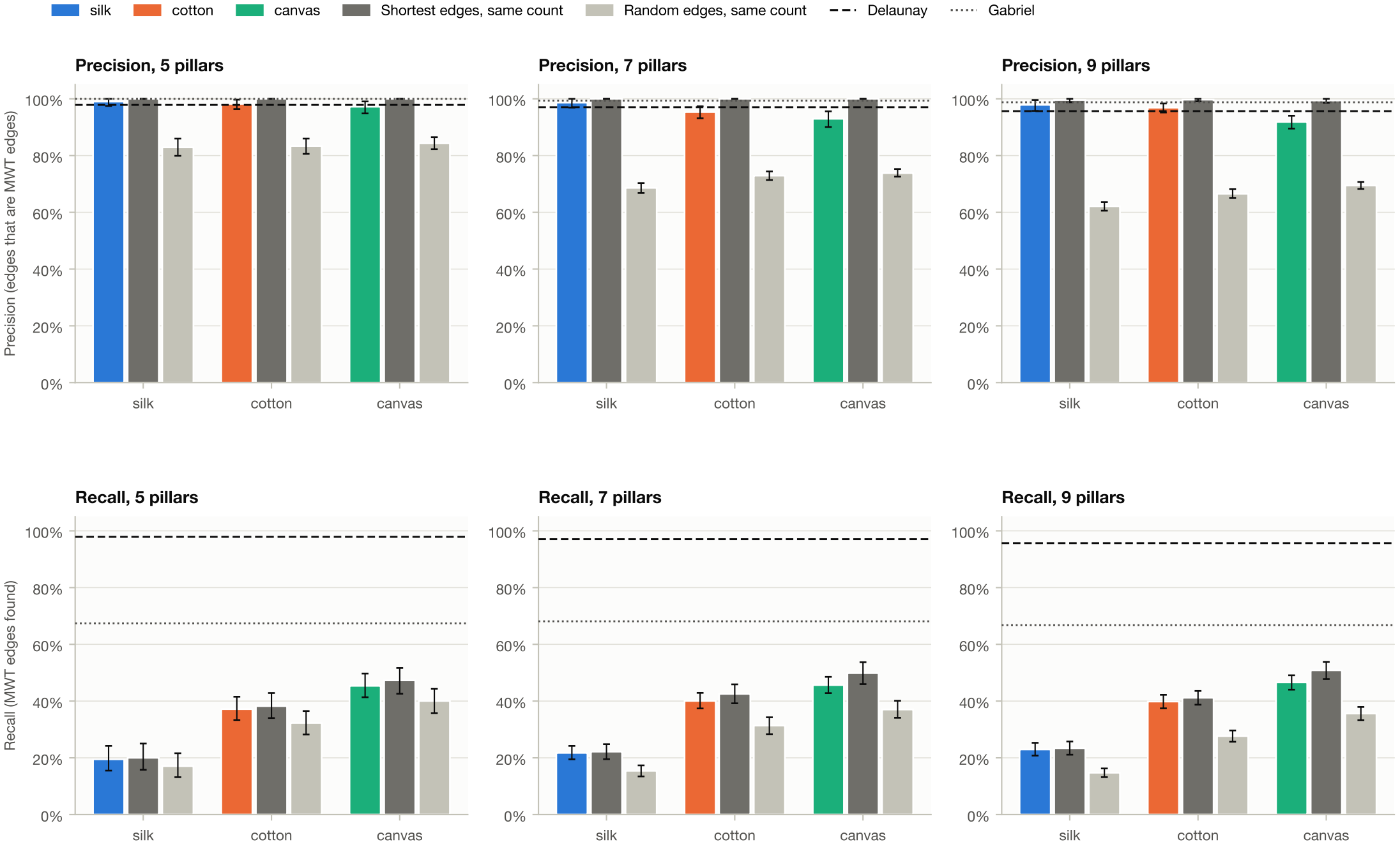
- silk: lighter / heavier than Greedy 0 / 2, than Delaunay 7 / 2
- cotton: lighter / heavier than Greedy 0 / 4, than Delaunay 6 / 4
- canvas: lighter / heavier than Greedy 0 / 6, than Delaunay 6 / 6

- silk: lighter / heavier than Greedy 1 / 3, than Delaunay 12 / 1
- cotton: lighter / heavier than Greedy 1 / 15, than Delaunay 11 / 13
- canvas: lighter / heavier than Greedy 1 / 19, than Delaunay 11 / 16

- silk: lighter / heavier than Greedy 1 / 4, than Delaunay 23 / 4
- cotton: lighter / heavier than Greedy 1 / 11, than Delaunay 23 / 8
- canvas: lighter / heavier than Greedy 0 / 29, than Delaunay 18 / 23

Is the cloth better than simple geometric rules?

Each cloth type against two rules that pick the SAME number of edges with no physics (shortest-first and random), plus two classic proximity graphs.

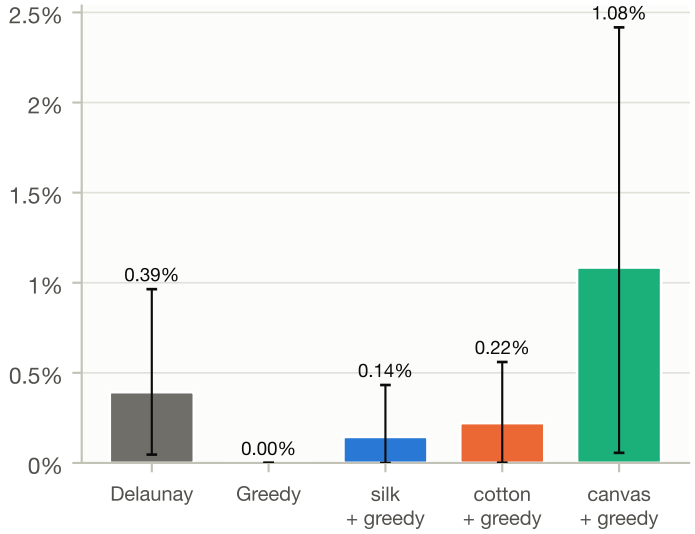


Which method gets closest to the MWT, without an exact search?

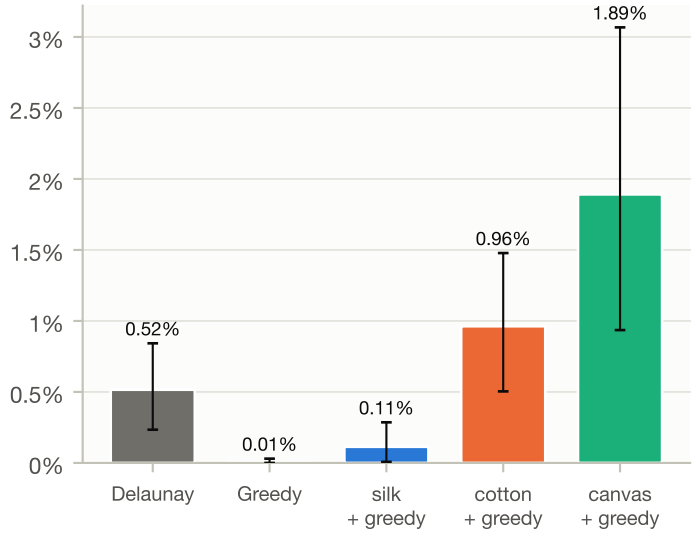
Gap = how much heavier the triangulation is than the minimum weight triangulation (lower is better). Cloth + greedy: the cloth's edges, completed shortest-first.



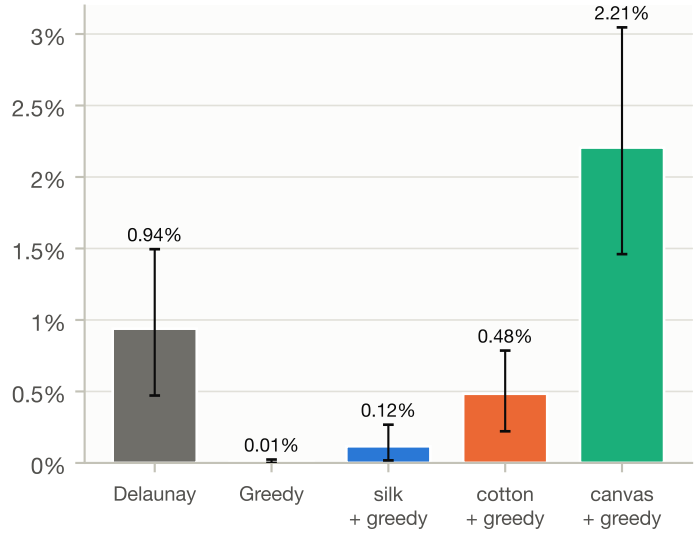
Mean gap above the MWT, 5 pillars



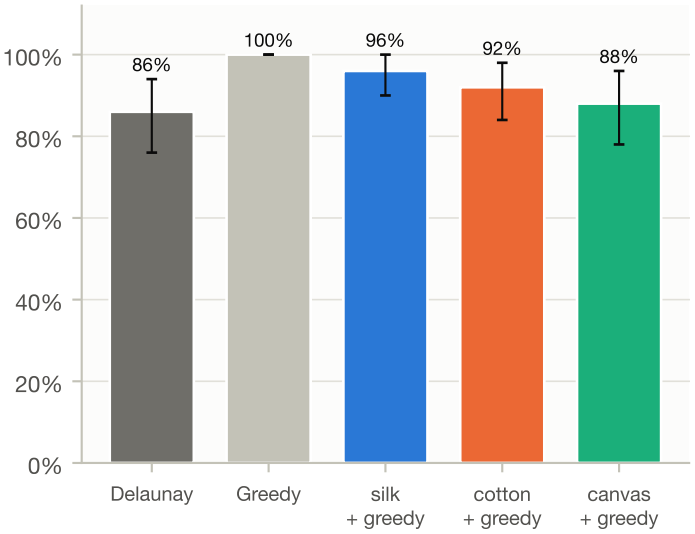
Mean gap above the MWT, 7 pillars



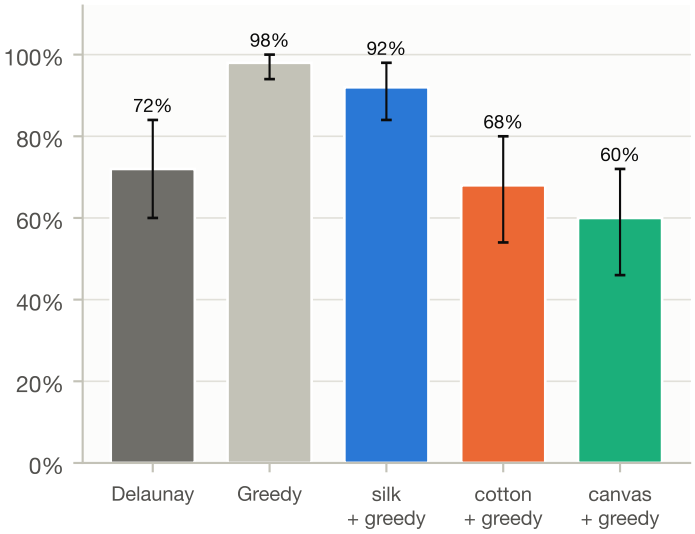
Mean gap above the MWT, 9 pillars



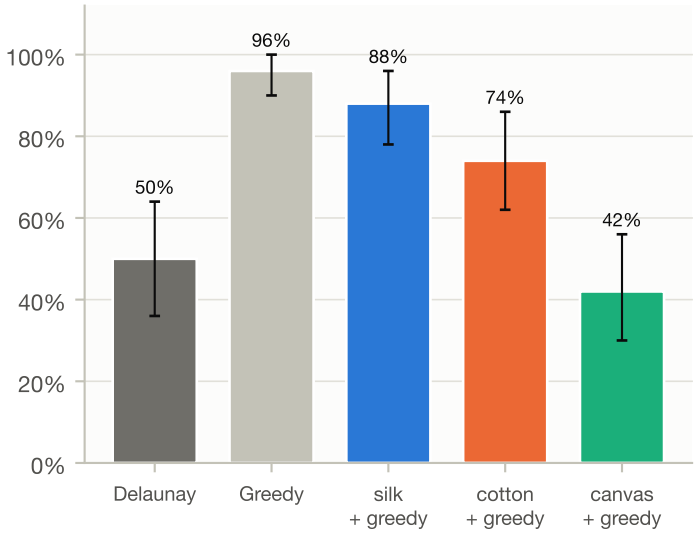
Trials that reach the exact MWT, 5 pillars



Trials that reach the exact MWT, 7 pillars



Trials that reach the exact MWT, 9 pillars

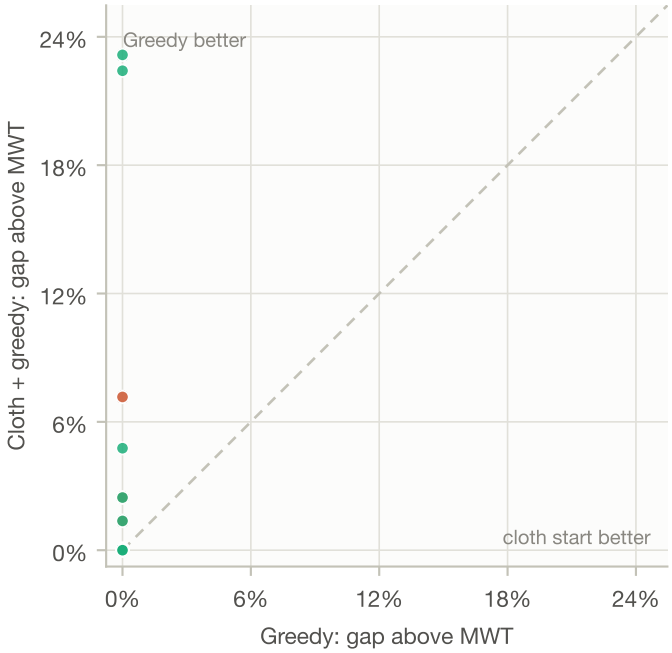


Does starting from the cloth beat plain Greedy?

silk cotton canvas

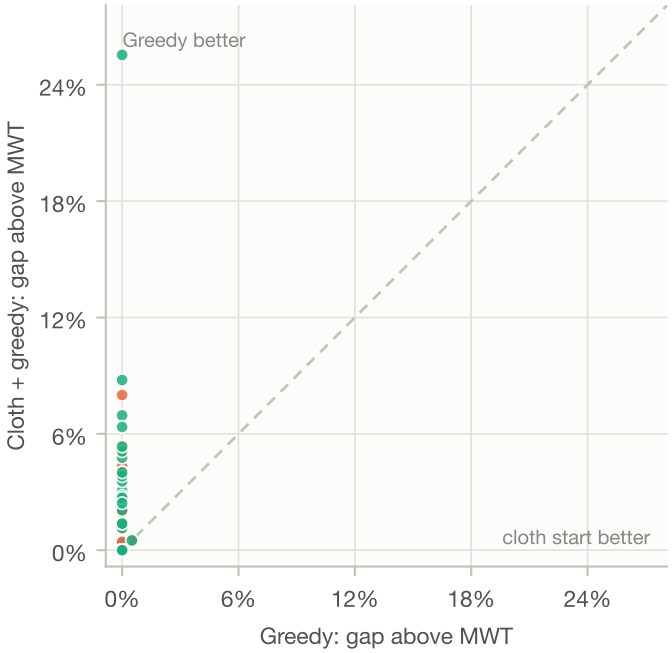
Each dot is one trial: Greedy's gap (across) against the cloth's edges + greedy completion (up), on the same layout. Below the diagonal the cloth start gave a lighter triangulation; on it, the same one.

5 pillars



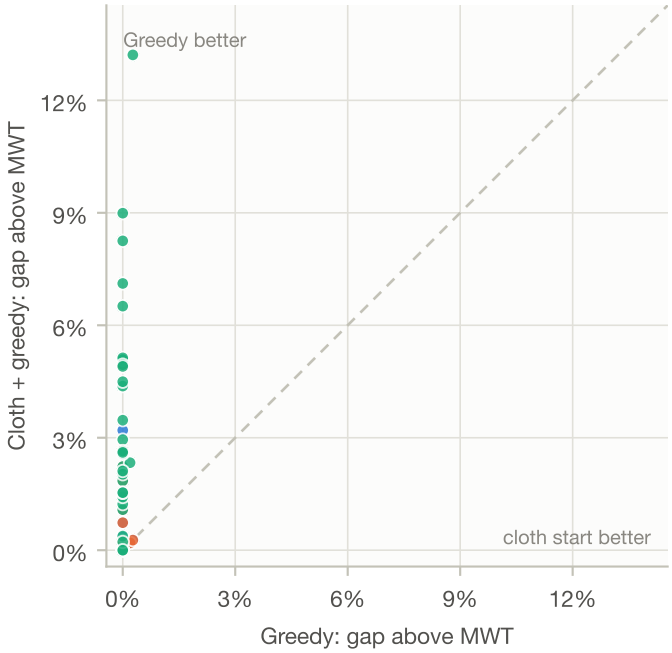
- silk: better in 0, same in 48, worse in 2 (p = 0.500)
- cotton: better in 0, same in 46, worse in 4 (p = 0.125)
- canvas: better in 0, same in 44, worse in 6 (p = 0.031)

7 pillars



- silk: better in 0, same in 47, worse in 3 (p = 0.250)
- cotton: better in 0, same in 35, worse in 15 (p < 0.001)
- canvas: better in 0, same in 31, worse in 19 (p < 0.001)

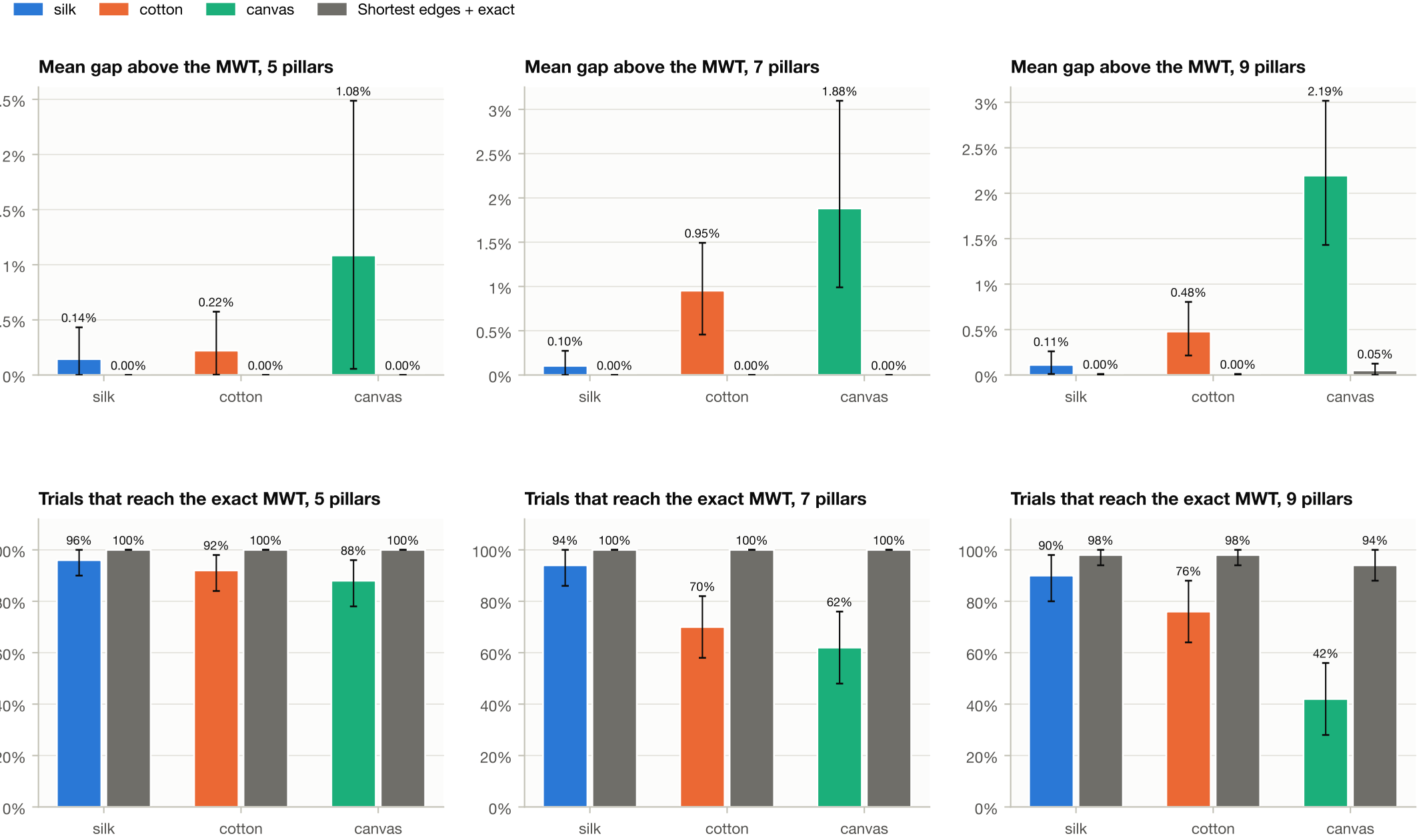
9 pillars



- silk: better in 0, same in 46, worse in 4 (p = 0.125)
- cotton: better in 0, same in 39, worse in 11 (p < 0.001)
- canvas: better in 0, same in 21, worse in 29 (p < 0.001)

With an exact completion: does the cloth's start lead to the MWT?

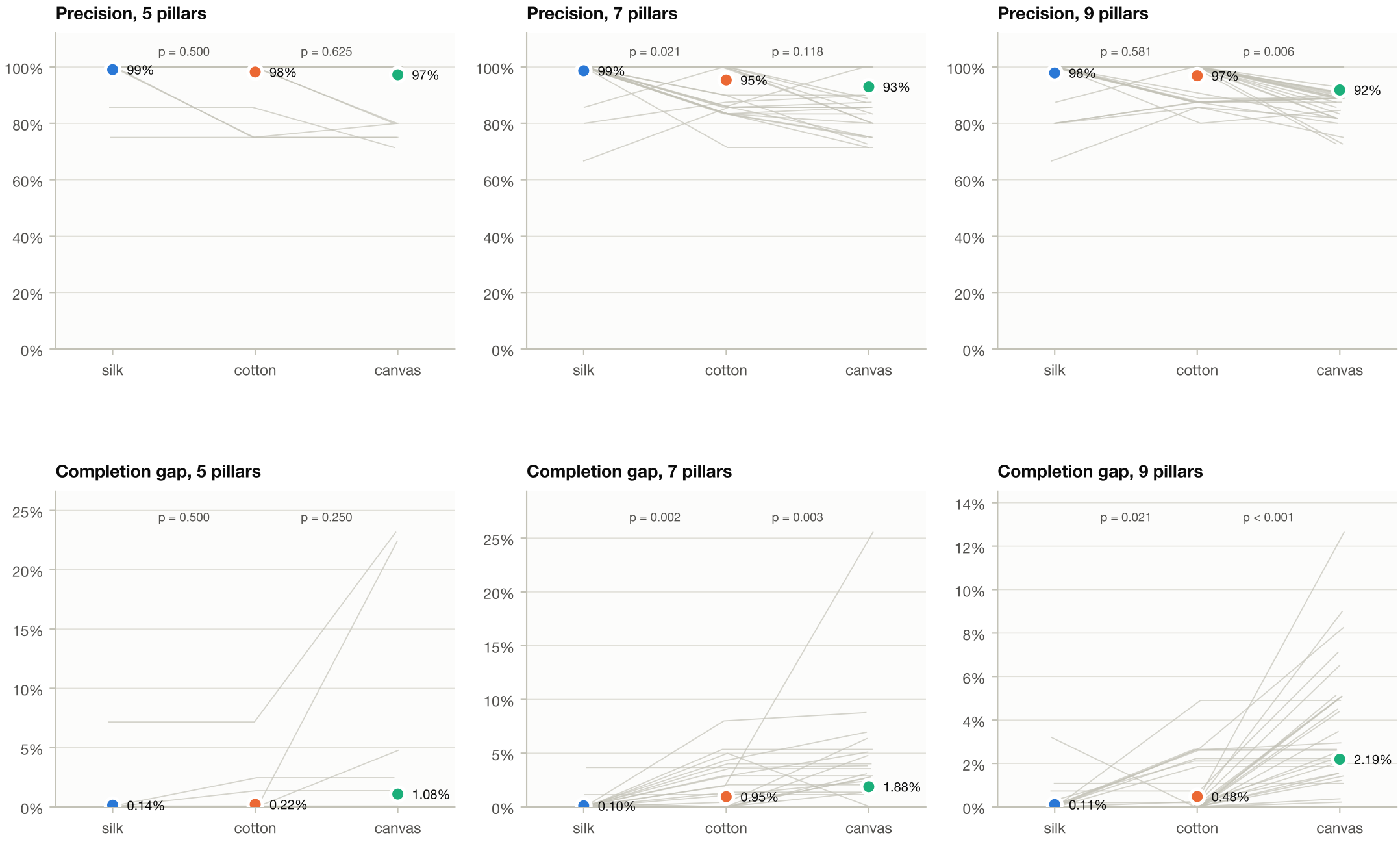
Both pipelines keep their starting edges and let the exact solver finish. Cloth: the cloth's own edges. Shortest: the same number of shortest edges, chosen without physics.



Search effort of the exact solver: 133 steps on average from scratch (most 1215), 43 starting from the cloth's edges, 46 starting from the same number of shortest edges. With 5 to 9 pillars the exact search is already instant, so these sizes cannot show a speed-up; only how close each start gets to the MWT.

Does the cloth type matter? Same layouts, different cloth

Each grey line is one layout dropped under every cloth type; coloured dots are the means. p-values: exact sign test on the paired differences between neighbouring cloth types.

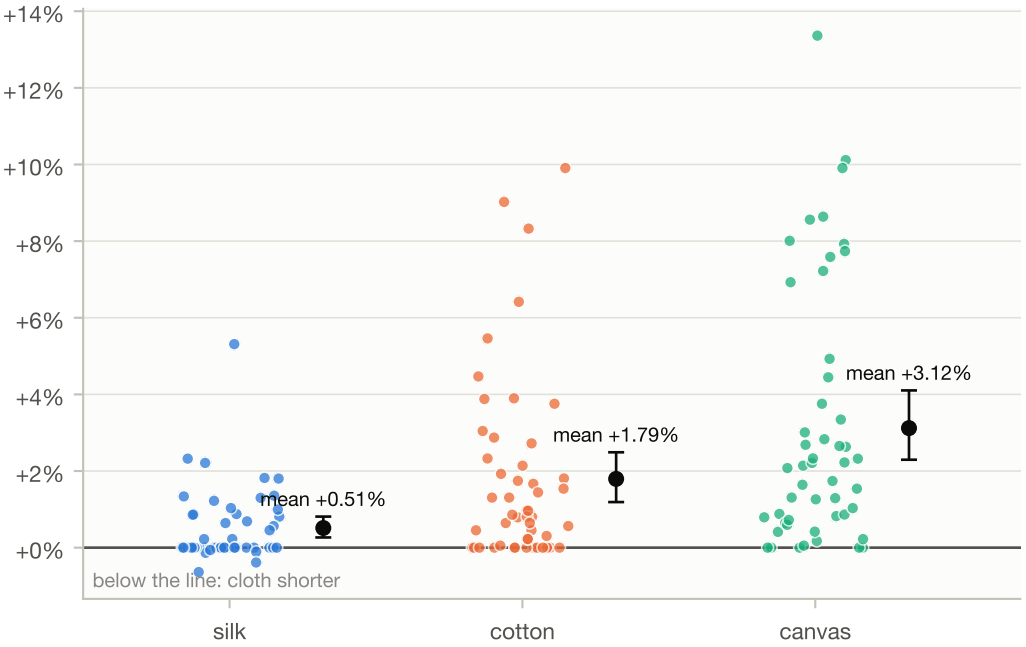


14 pillars: is the cloth's triangulation shorter than Greedy or Delaunay?

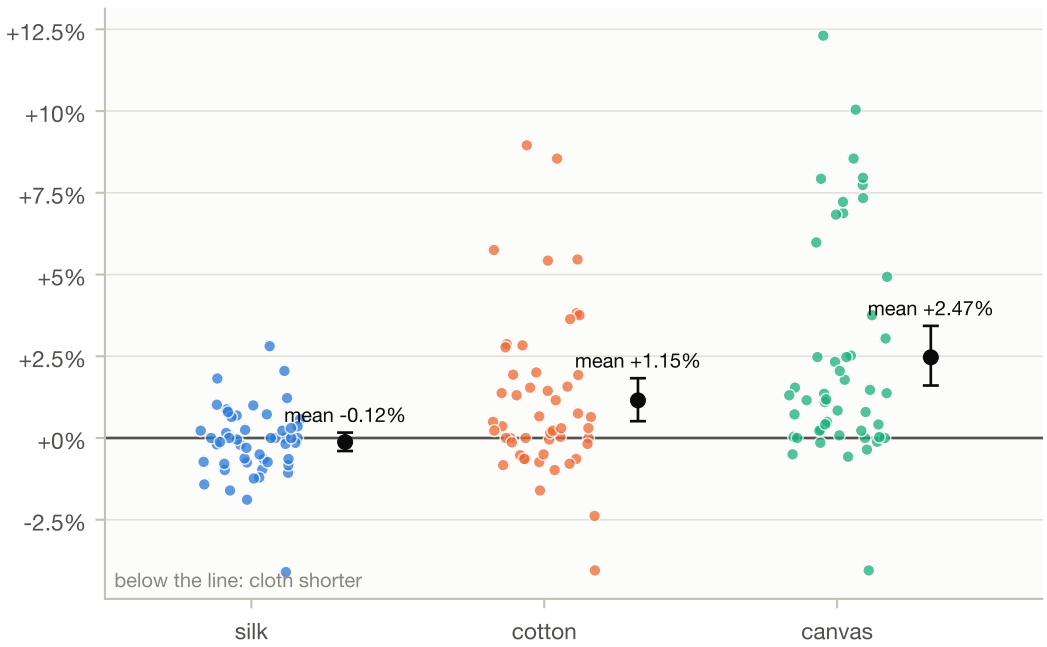
silk cotton canvas

Cloth = its own edges, completed by the exact search (the lightest triangulation keeping them). Greedy and Delaunay use the same pillar positions. Below 0% = the cloth's triangulation is shorter.

Cloth + exact completion vs Greedy, per trial



Cloth + exact completion vs Delaunay, per trial



	vs Greedy win / tie / loss	sign test	mean	vs Delaunay win / tie / loss	sign test	mean	cloth edges	of them Greedy's	completion proven
silk	5 / 23 / 22	p = 0.002	+0.51%	25 / 7 / 18	p = 0.360	-0.12%	10	95%	100%
cotton	0 / 13 / 37	p < 0.001	+1.79%	15 / 4 / 31	p = 0.026	+1.15%	15	92%	100%
canvas	0 / 5 / 45	p < 0.001	+3.12%	6 / 4 / 40	p < 0.001	+2.47%	19	90%	100%

Win / tie / loss: trials where the cloth's triangulation was shorter than, equal to or longer than the other method's on the same layout. Sign test: is shorter vs longer more lopsided than a coin flip? Fairness: the cloth side uses an exact search to finish its triangulation; Greedy and Delaunay use none. Completion proven: the search finished within Unity's node limit.

Key numbers

Generated from the data; read them together with the figures before drawing conclusions.

- Across all 450 trials, 96% of the edges the cloth formed were edges of the exact minimum weight triangulation (95% interval 96-97%).
- On average the cloth built 35% of the MWT by itself (34-37%); it formed a complete triangulation in 1% of trials.
- In 79% of trials every cloth edge could be completed into an exact MWT. The completion gap averaged 0.797% (median 0.000%, worst 25.54%).
- 5 pillars, precision by cloth type: silk 99%, cotton 98%, canvas 97%; inside an MWT: 92% of trials.
- 7 pillars, precision by cloth type: silk 99%, cotton 95%, canvas 93%; inside an MWT: 75% of trials.
- 9 pillars, precision by cloth type: silk 98%, cotton 97%, canvas 92%; inside an MWT: 69% of trials.
- Cloth vs. shortest-edges rule (same number of edges): the cloth's precision was higher in 0 trials, lower in 92, equal in 349 (mean difference -3.5 points, $p < 0.001$).
- Cloth vs. random edges (same number): higher in 413 trials, lower in 15 (mean difference +22.8 points, $p < 0.001$).
- For reference, the Delaunay triangulation shares 97% of its edges with the MWT and the Gabriel graph 99%.
- cotton vs. silk on the same layouts: precision higher in 8, lower in 23 ($p = 0.011$).
- canvas vs. cotton on the same layouts: precision higher in 11, lower in 35 ($p < 0.001$).
- Cloth + completion was exactly the MWT in: silk 93%, cotton 79%, canvas 64% of trials; Greedy in 98% of layouts, Delaunay in 69%. Against Greedy on the same layout, cloth + completion was lighter in 5 trials and heavier in 93.
- Whole triangulations without exact search, mean gap above the MWT: Delaunay 0.62%, Greedy 0.01%, cloth + greedy completion: silk 0.13%, cotton 0.56%, canvas 1.73%.
- Reaching the exact MWT without search: Delaunay in 69% of layouts, Greedy in 98%, cloth + greedy: silk 92%, cotton 78%, canvas 63%.
- silk + greedy vs. plain Greedy on the same layout: lighter in 0 trials, heavier in 9, the same in 141 ($p = 0.004$).
- cotton + greedy vs. plain Greedy on the same layout: lighter in 0 trials, heavier in 30, the same in 120 ($p < 0.001$).
- canvas + greedy vs. plain Greedy on the same layout: lighter in 0 trials, heavier in 54, the same in 96 ($p < 0.001$).

Key numbers (continued)

Generated from the data; read them together with the figures before drawing conclusions.

- With an exact completion, the start still reached the MWT in: cloth silk 93%, cotton 79%, canvas 64%; same number of shortest edges silk-sized 99%, cotton-sized 99%, canvas-sized 98%.
- Exact search effort: 133 steps on average from scratch, 43 from the cloth's edges and 46 from the same number of shortest edges (any fixed start shrinks the search). At 5 to 9 pillars the exact MWT is instant, so a speed-up from the cloth would need much larger point sets to show.
- Check: the MWT computed here matched Unity's MWT weight in 450 of 450 trials.
- Speed: a drop took 1.7 s of real time on average for 6.1 s of simulated time.
- 14 pillars, silk, cloth + exact completion: vs Greedy shorter in 5, same in 23, longer in 22 (mean +0.513%, $p = 0.002$); vs Delaunay shorter in 25, same in 7, longer in 18 (mean -0.124%, $p = 0.360$).
- 14 pillars, cotton, cloth + exact completion: vs Greedy shorter in 0, same in 13, longer in 37 (mean +1.794%, $p < 0.001$); vs Delaunay shorter in 15, same in 4, longer in 31 (mean +1.151%, $p = 0.026$).
- 14 pillars, canvas, cloth + exact completion: vs Greedy shorter in 0, same in 5, longer in 45 (mean +3.120%, $p < 0.001$); vs Delaunay shorter in 6, same in 4, longer in 40 (mean +2.468%, $p < 0.001$).
- 14 pillars: on average 92% of the cloth's edges were also Greedy edges.
- 14 pillars: the exact completion finished within Unity's node limit in 100% of trials. Note that it uses an exact search while Greedy and Delaunay use none.
- 5 pillars: the cloth's edges stopped changing in 150 of 150 trials, after 5.7 s of simulated time on average (0.9 changes along the way).
- 7 pillars: the cloth's edges stopped changing in 150 of 150 trials, after 6.1 s of simulated time on average (1.4 changes along the way).
- 9 pillars: the cloth's edges stopped changing in 150 of 150 trials, after 6.6 s of simulated time on average (2.0 changes along the way).
- 14 pillars: the cloth's edges stopped changing in 150 of 150 trials, after 8.6 s of simulated time on average (4.6 changes along the way).
- The cloth rested on a floor at Y = -1 below the pillars.

Key numbers (continued)

Generated from the data; read them together with the figures before drawing conclusions.

Cloth types

silk: bending 0.05, stretching 0.90, friction 0.30, damping 0.40

cotton: bending 0.40, stretching 1.00, friction 1.00, damping 0.60

canvas: bending 1.00, stretching 1.00, friction 1.00, damping 0.80

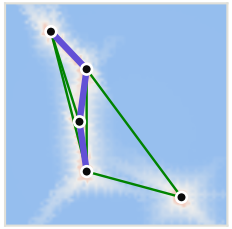
Cloth 20.0 x 20.0 units, 80 x 80 cells; analysed after 5.0 s simulated; pillars spread over 60% of the cloth width, at least 1.5 units apart.

First 30 of 50 trials: silk, 5 pillars

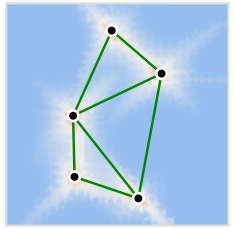
cloth edge in the MWT | cloth edge NOT in the MWT | MWT edge | cloth above threshold | cloth below threshold

50 layouts seen from above. Precision 99%, recall 19%, inside an MWT in 96% of trials, mean gap 0.145%.

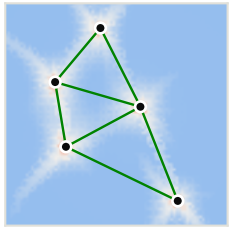
seed 6000 | precision 100% | gap 0.00%



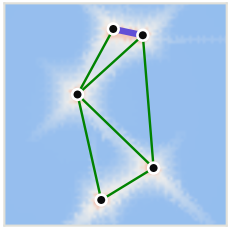
seed 6001 | precision - | gap 0.00%



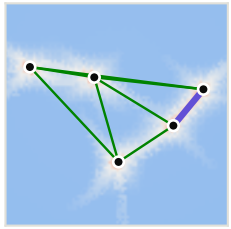
seed 6002 | precision - | gap 0.00%



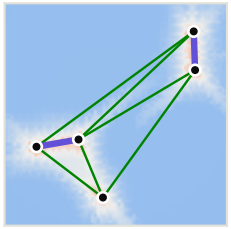
seed 6003 | precision 100% | gap 0.00%



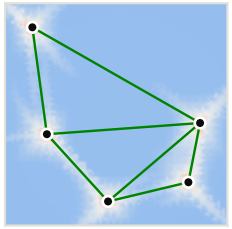
seed 6004 | precision 100% | gap 0.00%



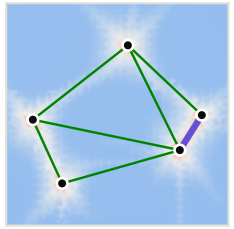
seed 6005 | precision 100% | gap 0.00%



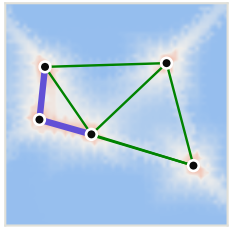
seed 6006 | precision - | gap 0.00%



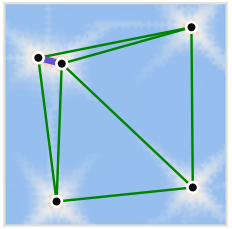
seed 6007 | precision 100% | gap 0.00%



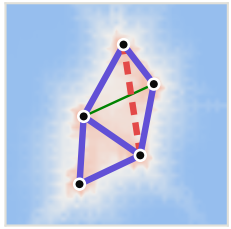
seed 6008 | precision 100% | gap 0.00%



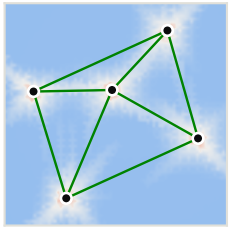
seed 6009 | precision 100% | gap 0.00%



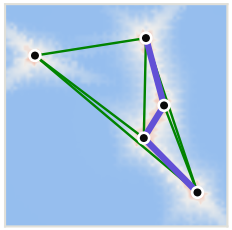
seed 6010 | precision 86% | gap 7.16%



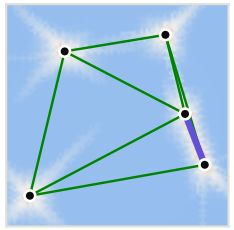
seed 6011 | precision - | gap 0.00%



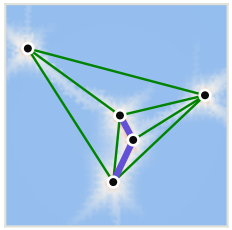
seed 6012 | precision 100% | gap 0.00%



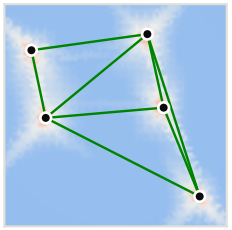
seed 6013 | precision 100% | gap 0.00%



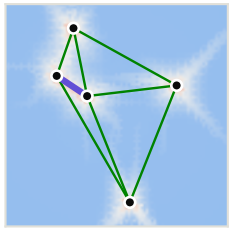
seed 6014 | precision 100% | gap 0.00%



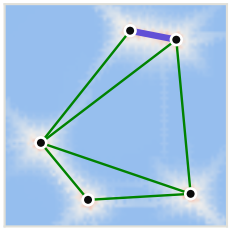
seed 6015 | precision - | gap 0.00%



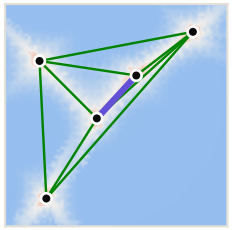
seed 6016 | precision 100% | gap 0.00%



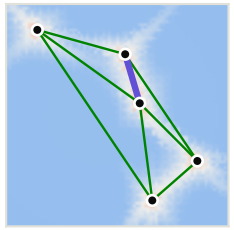
seed 6017 | precision 100% | gap 0.00%



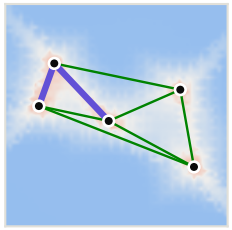
seed 6018 | precision 100% | gap 0.00%



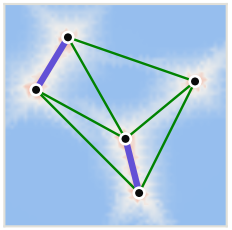
seed 6019 | precision 100% | gap 0.00%



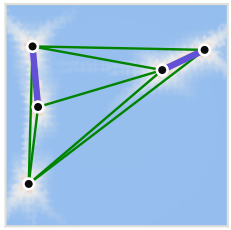
seed 6020 | precision 100% | gap 0.00%



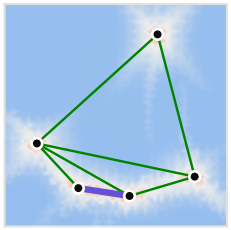
seed 6021 | precision 100% | gap 0.00%



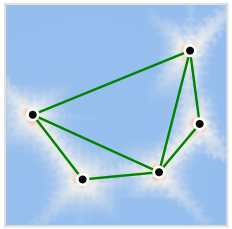
seed 6022 | precision 100% | gap 0.00%



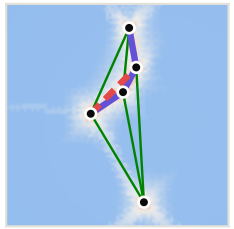
seed 6023 | precision 100% | gap 0.00%



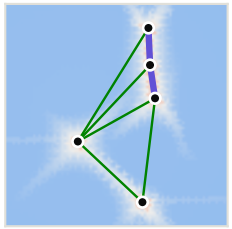
seed 6024 | precision - | gap 0.00%



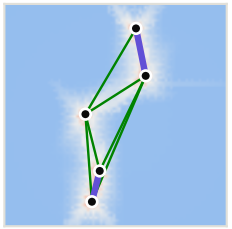
seed 6025 | precision 75% | gap 0.07%



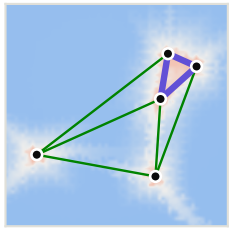
seed 6026 | precision 100% | gap 0.00%



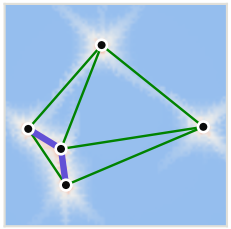
seed 6027 | precision 100% | gap 0.00%



seed 6028 | precision 100% | gap 0.00%



seed 6029 | precision 100% | gap 0.00%

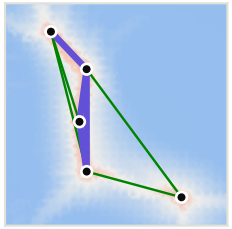


First 30 of 50 trials: cotton, 5 pillars

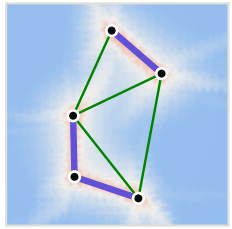
cloth edge in the MWT | cloth edge NOT in the MWT | MWT edge | cloth above threshold | cloth below threshold

50 layouts seen from above. Precision 98%, recall 37%, inside an MWT in 92% of trials, mean gap 0.221%.

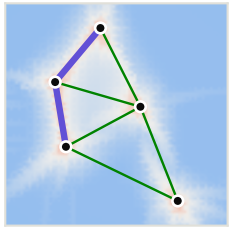
seed 6000 | precision 100% | gap 0.00%



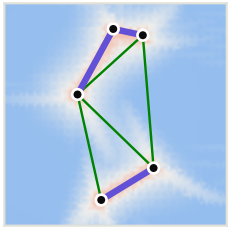
seed 6001 | precision 100% | gap 0.00%



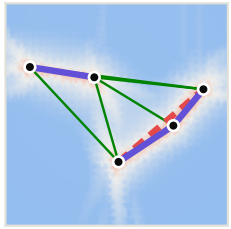
seed 6002 | precision 100% | gap 0.00%



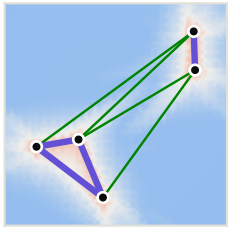
seed 6003 | precision 100% | gap 0.00%



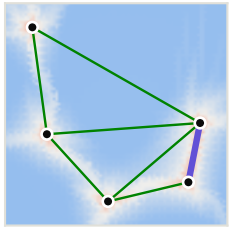
seed 6004 | precision 75% | gap 2.46%



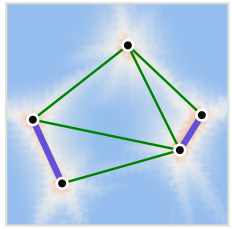
seed 6005 | precision 100% | gap 0.00%



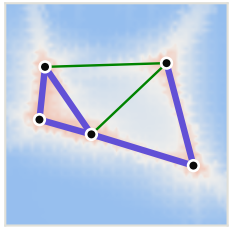
seed 6006 | precision 100% | gap 0.00%



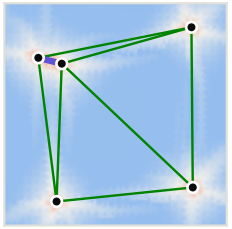
seed 6007 | precision 100% | gap 0.00%



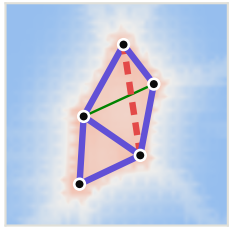
seed 6008 | precision 100% | gap 0.00%



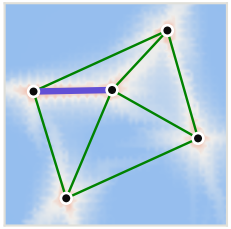
seed 6009 | precision 100% | gap 0.00%



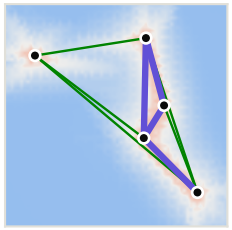
seed 6010 | precision 86% | gap 7.16%



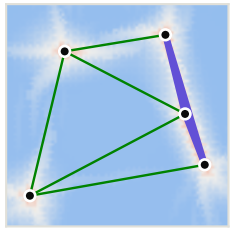
seed 6011 | precision 100% | gap 0.00%



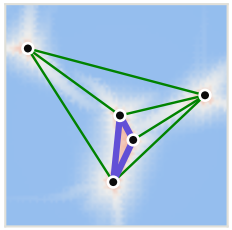
seed 6012 | precision 100% | gap 0.00%



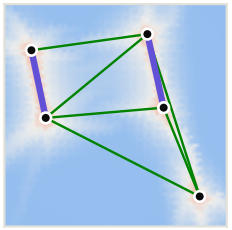
seed 6013 | precision 100% | gap 0.00%



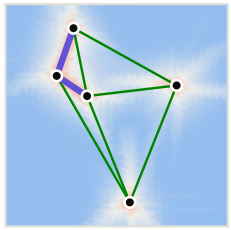
seed 6014 | precision 100% | gap 0.00%



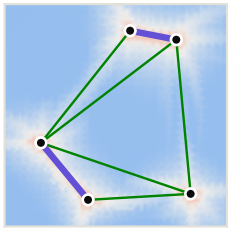
seed 6015 | precision 100% | gap 0.00%



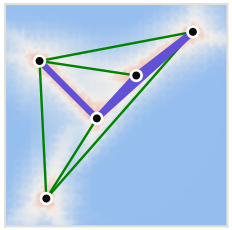
seed 6016 | precision 100% | gap 0.00%



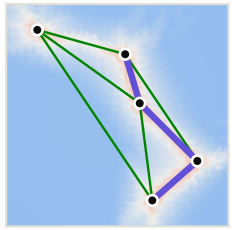
seed 6017 | precision 100% | gap 0.00%



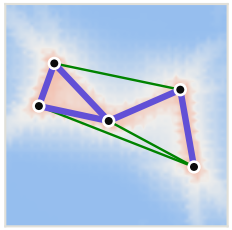
seed 6018 | precision 100% | gap 0.00%



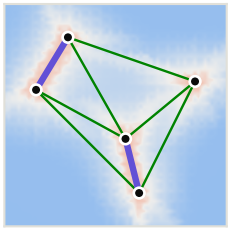
seed 6019 | precision 100% | gap 0.00%



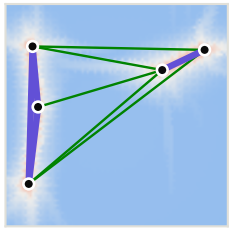
seed 6020 | precision 100% | gap 0.00%



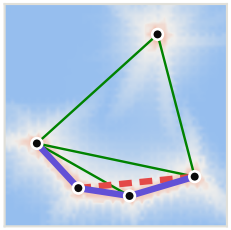
seed 6021 | precision 100% | gap 0.00%



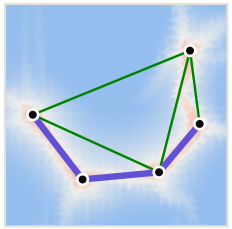
seed 6022 | precision 100% | gap 0.00%



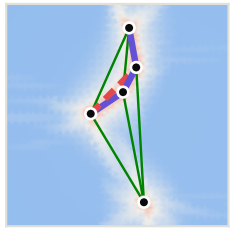
seed 6023 | precision 75% | gap 1.37%



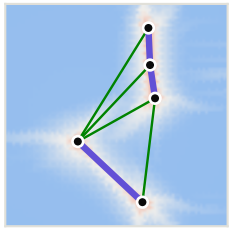
seed 6024 | precision 100% | gap 0.00%



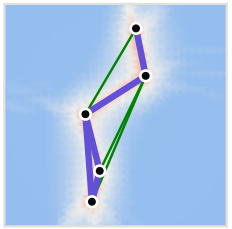
seed 6025 | precision 75% | gap 0.07%



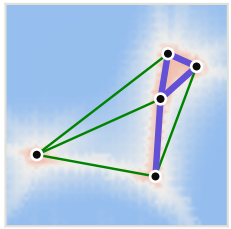
seed 6026 | precision 100% | gap 0.00%



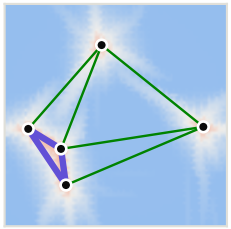
seed 6027 | precision 100% | gap 0.00%



seed 6028 | precision 100% | gap 0.00%



seed 6029 | precision 100% | gap 0.00%

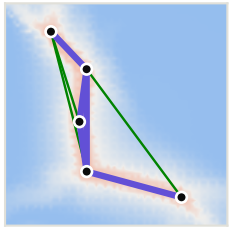


First 30 of 50 trials: canvas, 5 pillars

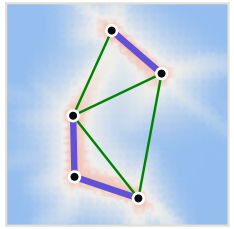
cloth edge in the MWT | cloth edge NOT in the MWT | MWT edge | cloth above threshold | cloth below threshold

50 layouts seen from above. Precision 97%, recall 45%, inside an MWT in 88% of trials, mean gap 1.085%.

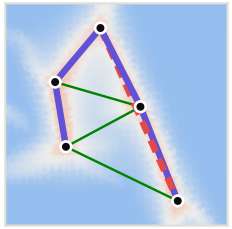
seed 6000 | precision 100% | gap 0.00%



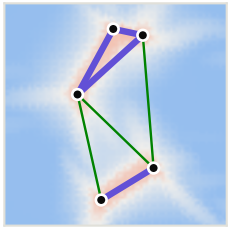
seed 6001 | precision 100% | gap 0.00%



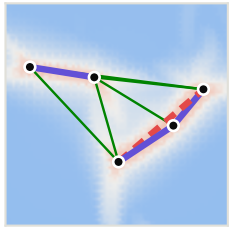
seed 6002 | precision 80% | gap 22.41%



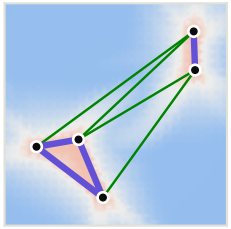
seed 6003 | precision 100% | gap 0.00%



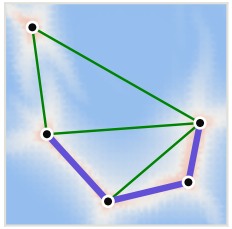
seed 6004 | precision 75% | gap 2.46%



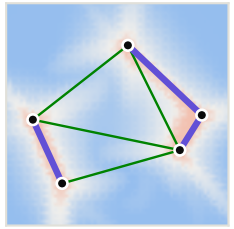
seed 6005 | precision 100% | gap 0.00%



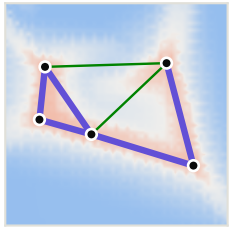
seed 6006 | precision 100% | gap 0.00%



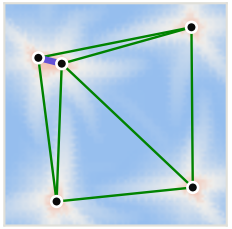
seed 6007 | precision 100% | gap 0.00%



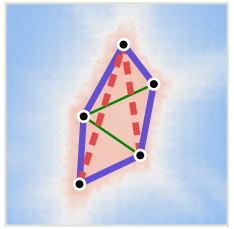
seed 6008 | precision 100% | gap 0.00%



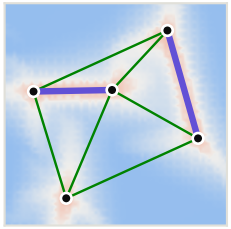
seed 6009 | precision 100% | gap 0.00%



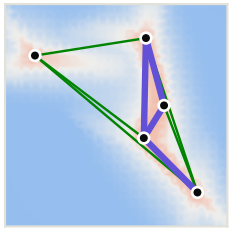
seed 6010 | precision 71% | gap 23.16%



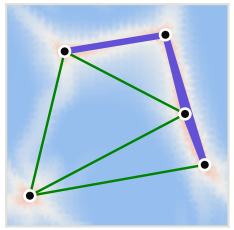
seed 6011 | precision 100% | gap 0.00%



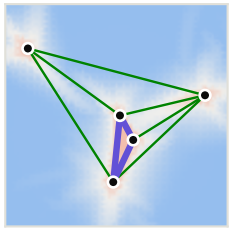
seed 6012 | precision 100% | gap 0.00%



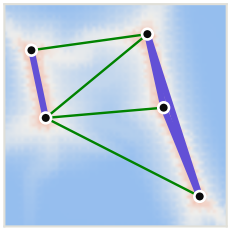
seed 6013 | precision 100% | gap 0.00%



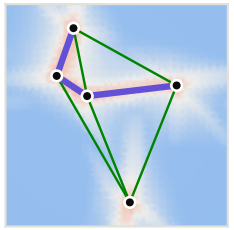
seed 6014 | precision 100% | gap 0.00%



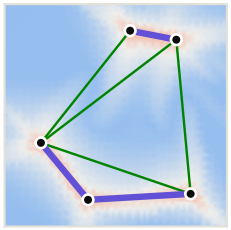
seed 6015 | precision 100% | gap 0.00%



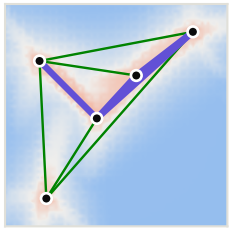
seed 6016 | precision 100% | gap 0.00%



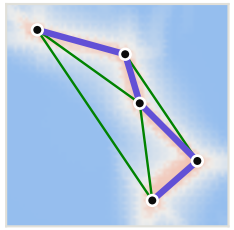
seed 6017 | precision 100% | gap 0.00%



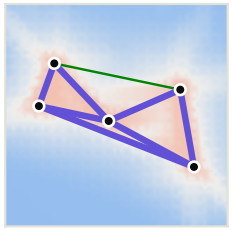
seed 6018 | precision 100% | gap 0.00%



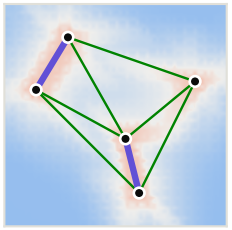
seed 6019 | precision 100% | gap 0.00%



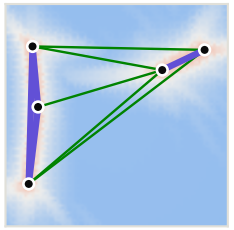
seed 6020 | precision 100% | gap 0.00%



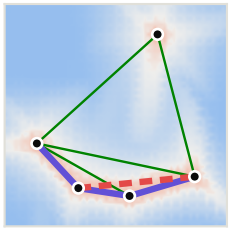
seed 6021 | precision 100% | gap 0.00%



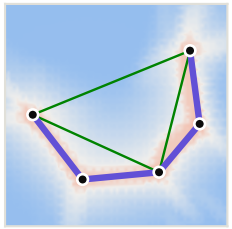
seed 6022 | precision 100% | gap 0.00%



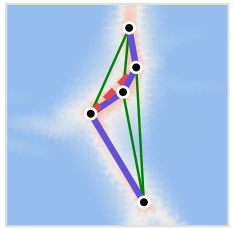
seed 6023 | precision 75% | gap 1.37%



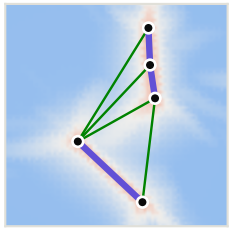
seed 6024 | precision 100% | gap 0.00%



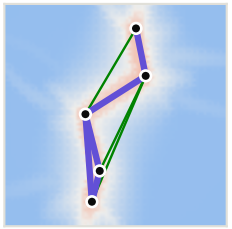
seed 6025 | precision 80% | gap 0.07%



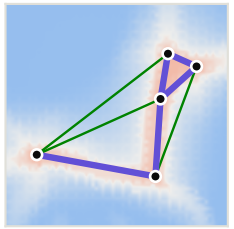
seed 6026 | precision 100% | gap 0.00%



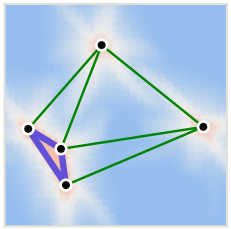
seed 6027 | precision 100% | gap 0.00%



seed 6028 | precision 100% | gap 0.00%



seed 6029 | precision 100% | gap 0.00%

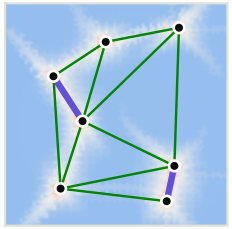


First 30 of 50 trials: silk, 7 pillars

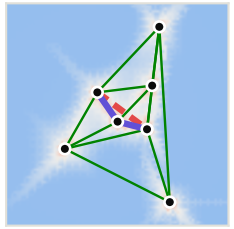
cloth edge in the MWT | cloth edge NOT in the MWT | MWT edge | cloth above threshold | cloth below threshold

50 layouts seen from above. Precision 99%, recall 22%, inside an MWT in 94% of trials, mean gap 0.104%.

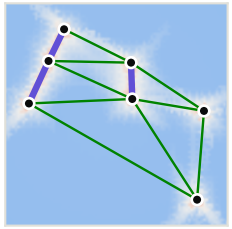
seed 8000 | precision 100% | gap 0.00%



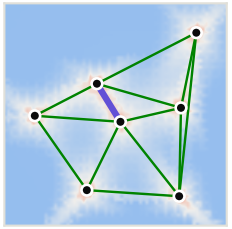
seed 8001 | precision 67% | gap 1.15%



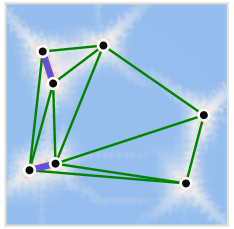
seed 8002 | precision 100% | gap 0.00%



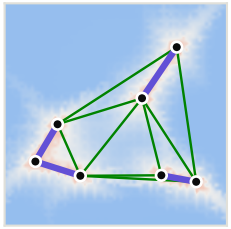
seed 8003 | precision 100% | gap 0.00%



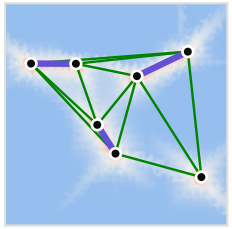
seed 8004 | precision 100% | gap 0.00%



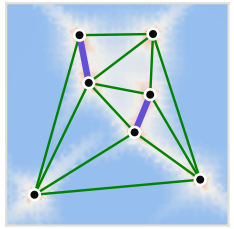
seed 8005 | precision 100% | gap 0.00%



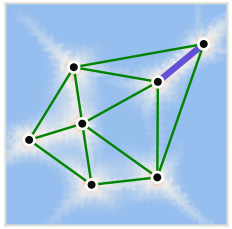
seed 8006 | precision 100% | gap 0.00%



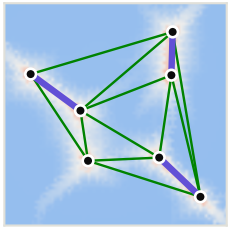
seed 8007 | precision 100% | gap 0.00%



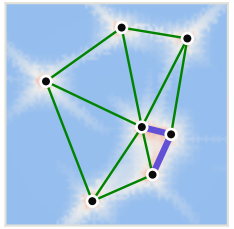
seed 8008 | precision 100% | gap 0.00%



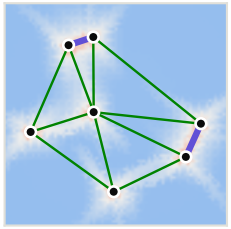
seed 8009 | precision 100% | gap 0.00%



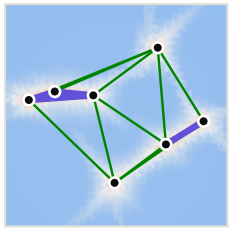
seed 8010 | precision 100% | gap 0.00%



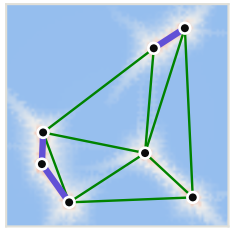
seed 8011 | precision 100% | gap 0.00%



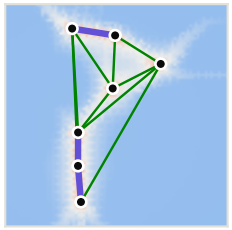
seed 8012 | precision 100% | gap 0.00%



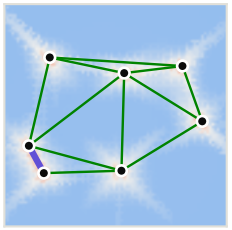
seed 8013 | precision 100% | gap 0.00%



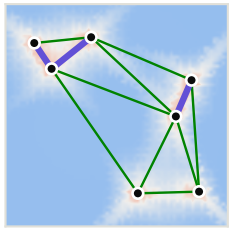
seed 8014 | precision 100% | gap 0.00%



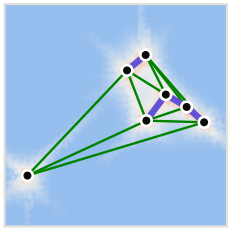
seed 8015 | precision 100% | gap 0.00%



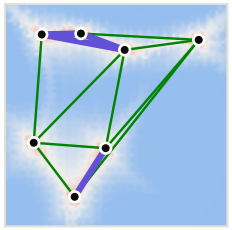
seed 8016 | precision 100% | gap 0.00%



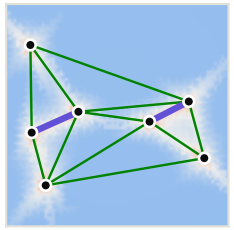
seed 8017 | precision 100% | gap 0.00%



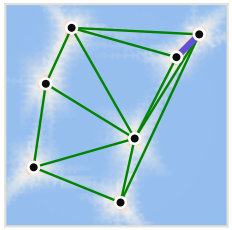
seed 8018 | precision 100% | gap 0.00%



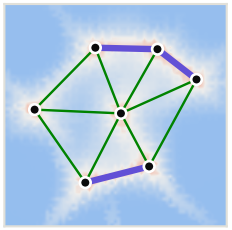
seed 8019 | precision 100% | gap 0.00%



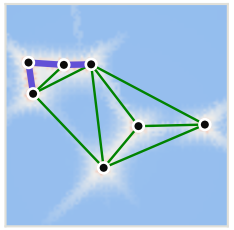
seed 8020 | precision 100% | gap 0.00%



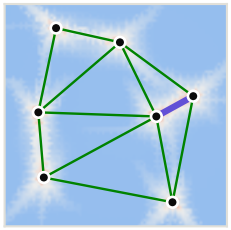
seed 8021 | precision 100% | gap 0.00%



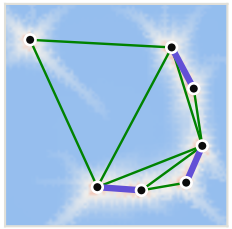
seed 8022 | precision 100% | gap 0.00%



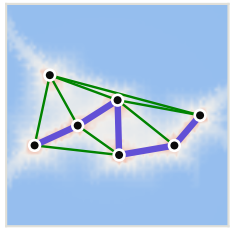
seed 8023 | precision 100% | gap 0.00%



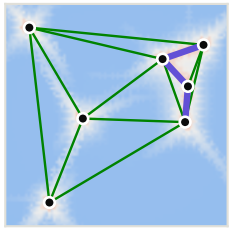
seed 8024 | precision 100% | gap 0.00%



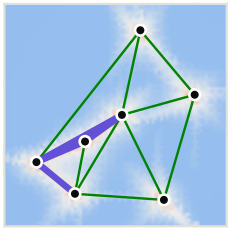
seed 8025 | precision 100% | gap 0.00%



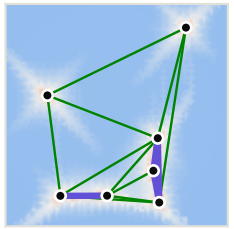
seed 8026 | precision 100% | gap 0.00%



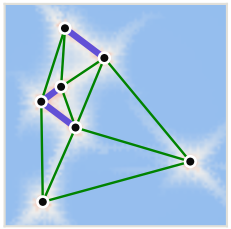
seed 8027 | precision 100% | gap 0.00%



seed 8028 | precision 100% | gap 0.00%



seed 8029 | precision 100% | gap 0.00%

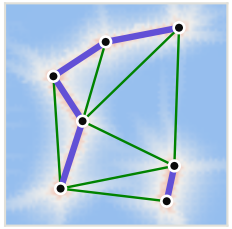


First 30 of 50 trials: cotton, 7 pillars

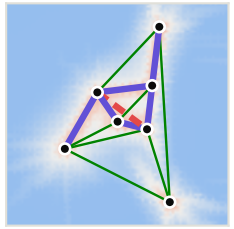
cloth edge in the MWT | cloth edge NOT in the MWT | MWT edge | cloth above threshold | cloth below threshold

50 layouts seen from above. Precision 95%, recall 40%, inside an MWT in 70% of trials, mean gap 0.953%.

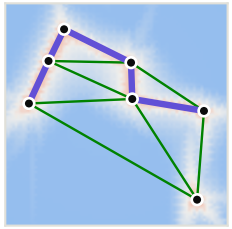
seed 8000 | precision 100% | gap 0.00%



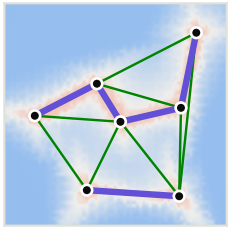
seed 8001 | precision 86% | gap 1.15%



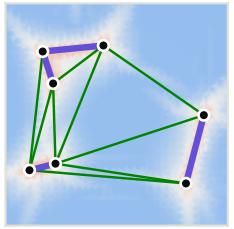
seed 8002 | precision 100% | gap 0.00%



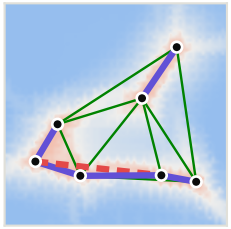
seed 8003 | precision 100% | gap 0.00%



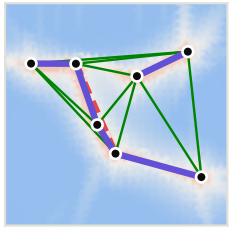
seed 8004 | precision 100% | gap 0.00%



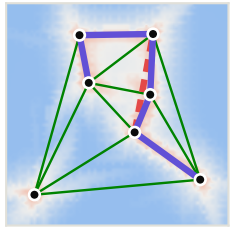
seed 8005 | precision 83% | gap 8.01%



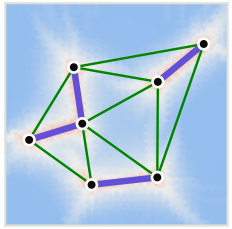
seed 8006 | precision 83% | gap 2.89%



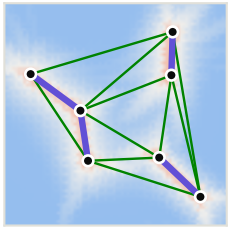
seed 8007 | precision 83% | gap 2.79%



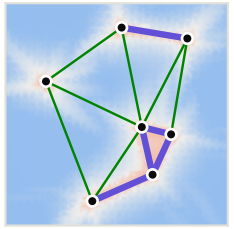
seed 8008 | precision 100% | gap 0.00%



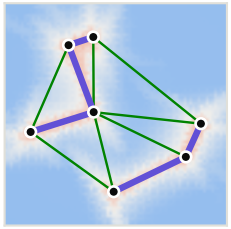
seed 8009 | precision 100% | gap 0.00%



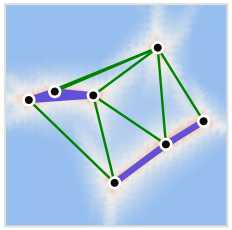
seed 8010 | precision 100% | gap 0.00%



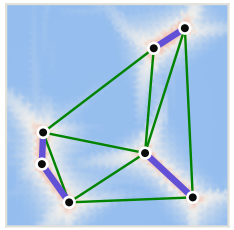
seed 8011 | precision 100% | gap 0.00%



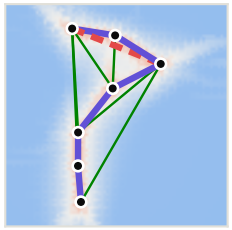
seed 8012 | precision 100% | gap 0.00%



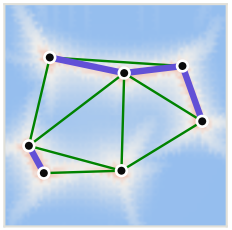
seed 8013 | precision 100% | gap 0.00%



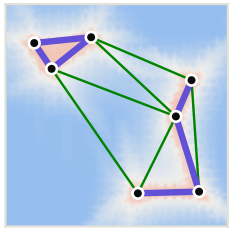
seed 8014 | precision 86% | gap 4.32%



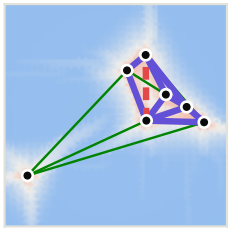
seed 8015 | precision 100% | gap 0.00%



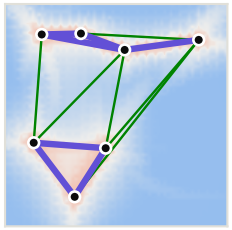
seed 8016 | precision 100% | gap 0.00%



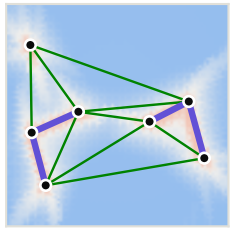
seed 8017 | precision 90% | gap 2.06%



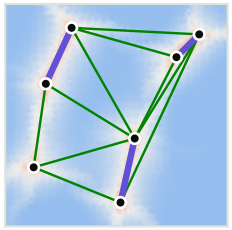
seed 8018 | precision 100% | gap 0.00%



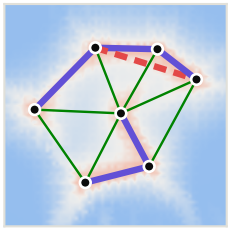
seed 8019 | precision 100% | gap 0.00%



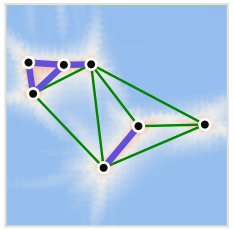
seed 8020 | precision 100% | gap 0.00%



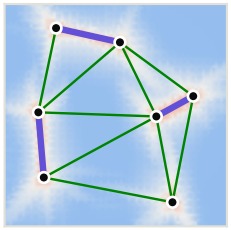
seed 8021 | precision 83% | gap 3.56%



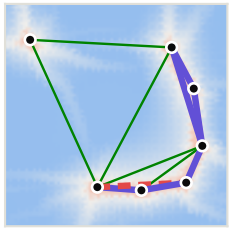
seed 8022 | precision 100% | gap 0.00%



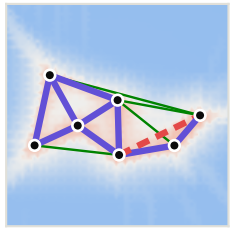
seed 8023 | precision 100% | gap 0.00%



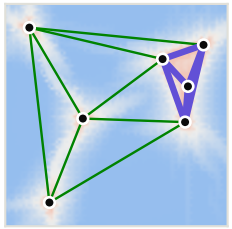
seed 8024 | precision 86% | gap 1.37%



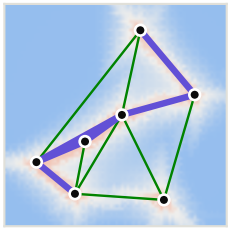
seed 8025 | precision 90% | gap 1.95%



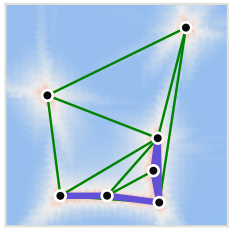
seed 8026 | precision 100% | gap 0.00%



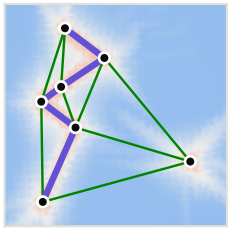
seed 8027 | precision 100% | gap 0.00%



seed 8028 | precision 100% | gap 0.00%



seed 8029 | precision 100% | gap 0.00%

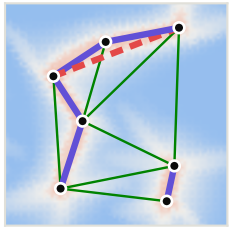


First 30 of 50 trials: canvas, 7 pillars

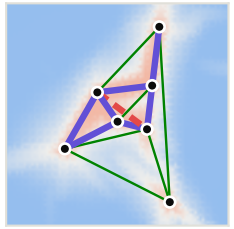
cloth edge in the MWT | cloth edge NOT in the MWT | MWT edge | cloth above threshold | cloth below threshold

50 layouts seen from above. Precision 93%, recall 46%, inside an MWT in 62% of trials, mean gap 1.881%.

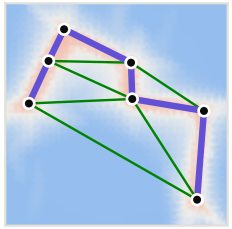
seed 8000 | precision 83% | gap 4.78%



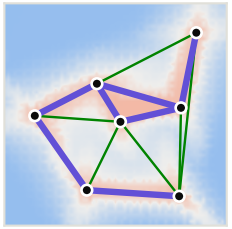
seed 8001 | precision 88% | gap 1.15%



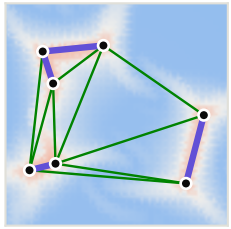
seed 8002 | precision 100% | gap 0.00%



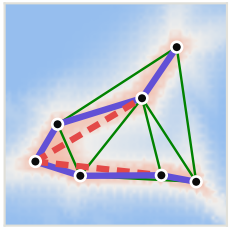
seed 8003 | precision 100% | gap 0.00%



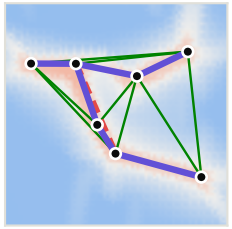
seed 8004 | precision 100% | gap 0.00%



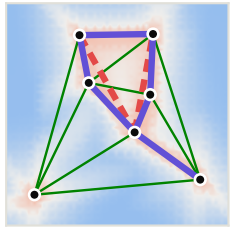
seed 8005 | precision 75% | gap 8.77%



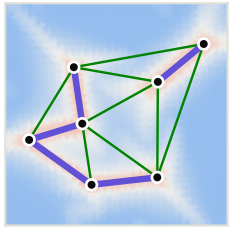
seed 8006 | precision 86% | gap 2.89%



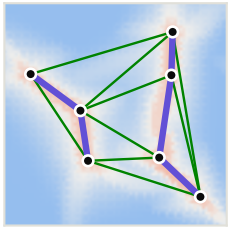
seed 8007 | precision 75% | gap 5.10%



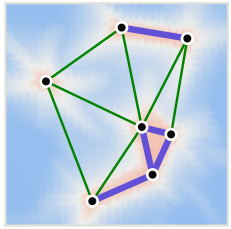
seed 8008 | precision 100% | gap 0.00%



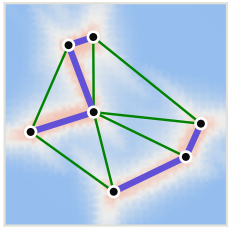
seed 8009 | precision 100% | gap 0.00%



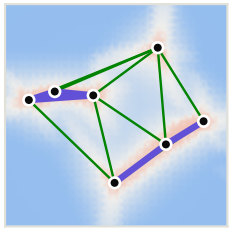
seed 8010 | precision 100% | gap 0.00%



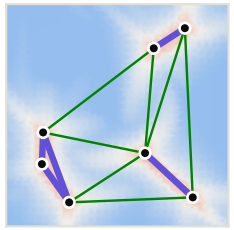
seed 8011 | precision 100% | gap 0.00%



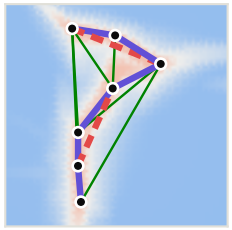
seed 8012 | precision 100% | gap 0.00%



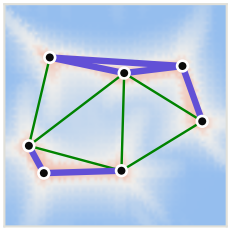
seed 8013 | precision 100% | gap 0.00%



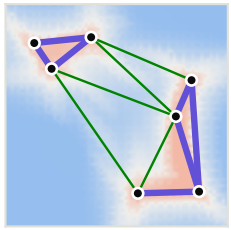
seed 8014 | precision 75% | gap 6.95%



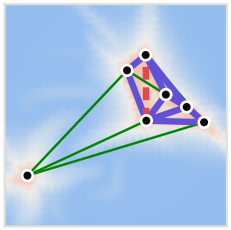
seed 8015 | precision 100% | gap 0.00%



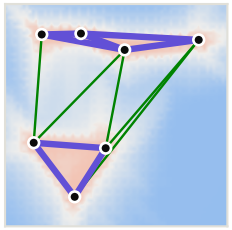
seed 8016 | precision 100% | gap 0.00%



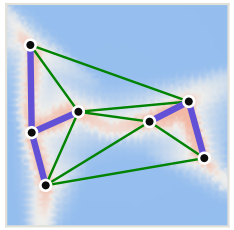
seed 8017 | precision 90% | gap 2.06%



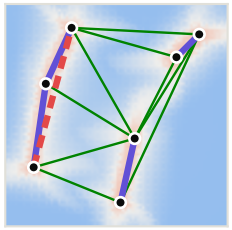
seed 8018 | precision 100% | gap 0.00%



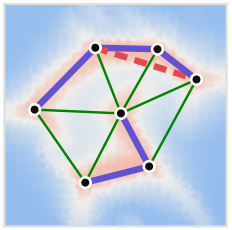
seed 8019 | precision 100% | gap 0.00%



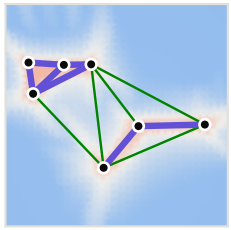
seed 8020 | precision 80% | gap 3.07%



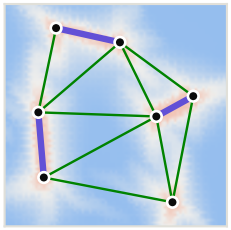
seed 8021 | precision 83% | gap 3.56%



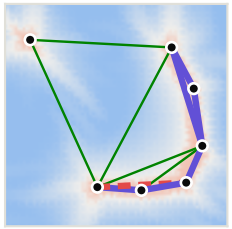
seed 8022 | precision 100% | gap 0.00%



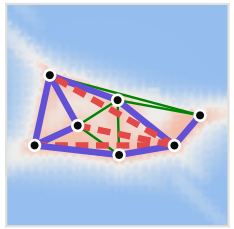
seed 8023 | precision 100% | gap 0.00%



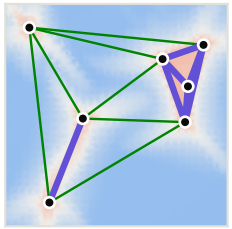
seed 8024 | precision 86% | gap 1.37%



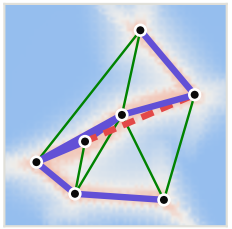
seed 8025 | precision 73% | gap 25.54%



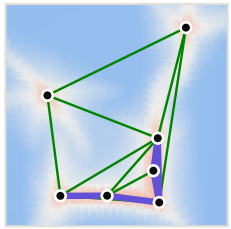
seed 8026 | precision 100% | gap 0.00%



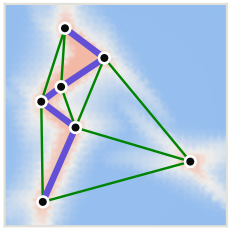
seed 8027 | precision 88% | gap 2.80%



seed 8028 | precision 100% | gap 0.00%



seed 8029 | precision 100% | gap 0.00%

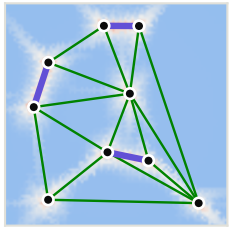


First 30 of 50 trials: silk, 9 pillars

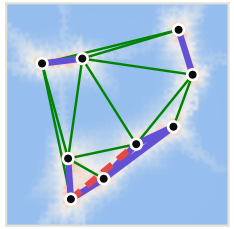
cloth edge in the MWT | cloth edge NOT in the MWT | MWT edge | cloth above threshold | cloth below threshold

50 layouts seen from above. Precision 98%, recall 23%, inside an MWT in 90% of trials, mean gap 0.111%.

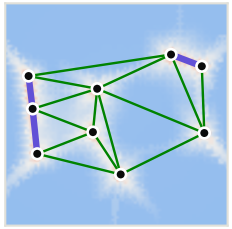
seed 10000 | precision 100% | gap 0.00%



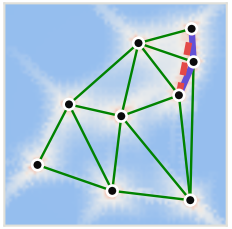
seed 10001 | precision 88% | gap 3.20%



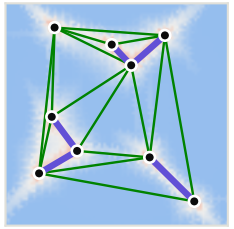
seed 10002 | precision 100% | gap 0.00%



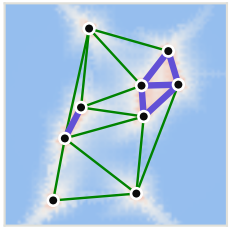
seed 10003 | precision 67% | gap 0.73%



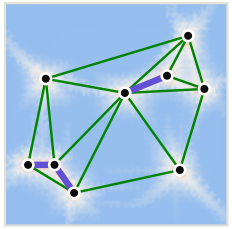
seed 10004 | precision 100% | gap 0.00%



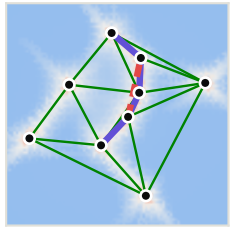
seed 10005 | precision 100% | gap 0.00%



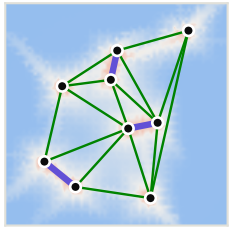
seed 10006 | precision 100% | gap 0.00%



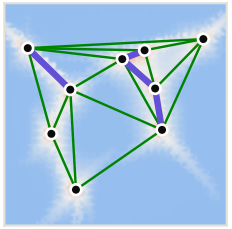
seed 10007 | precision 80% | gap 0.20%



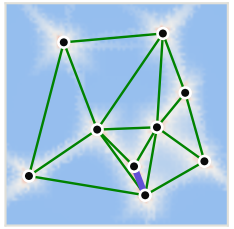
seed 10008 | precision 100% | gap 0.00%



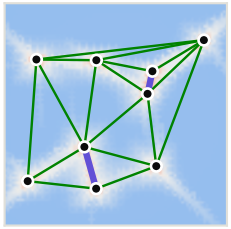
seed 10009 | precision 100% | gap 0.00%



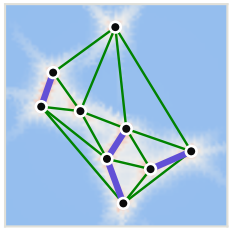
seed 10010 | precision 100% | gap 0.00%



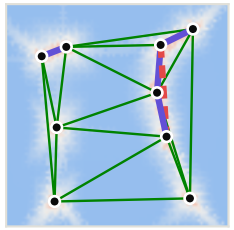
seed 10011 | precision 100% | gap 0.00%



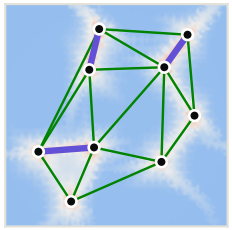
seed 10012 | precision 100% | gap 0.00%



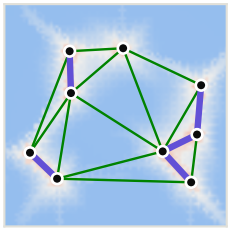
seed 10013 | precision 80% | gap 1.08%



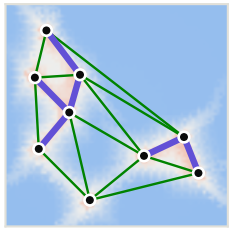
seed 10014 | precision 100% | gap 0.00%



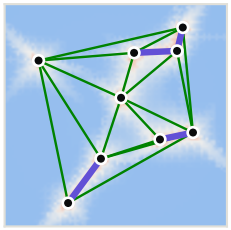
seed 10015 | precision 100% | gap 0.00%



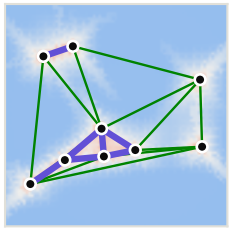
seed 10016 | precision 100% | gap 0.00%



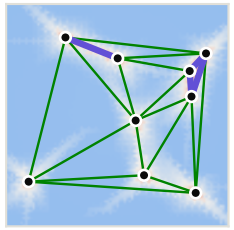
seed 10017 | precision 100% | gap 0.00%



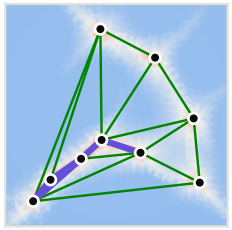
seed 10018 | precision 100% | gap 0.00%



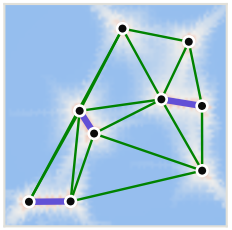
seed 10019 | precision 100% | gap 0.00%



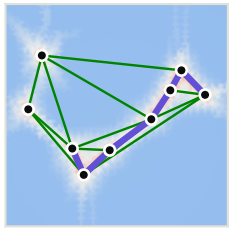
seed 10020 | precision 100% | gap 0.00%



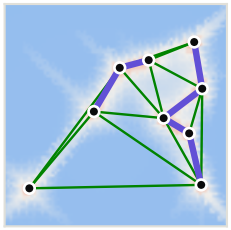
seed 10021 | precision 100% | gap 0.00%



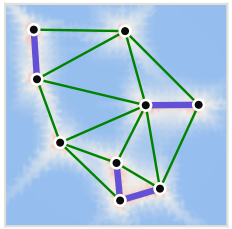
seed 10022 | precision 100% | gap 0.00%



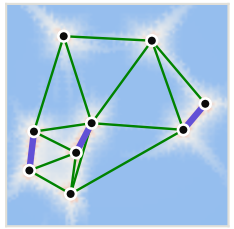
seed 10023 | precision 100% | gap 0.00%



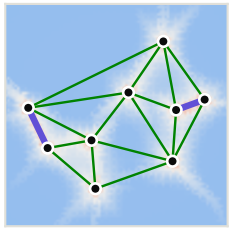
seed 10024 | precision 100% | gap 0.00%



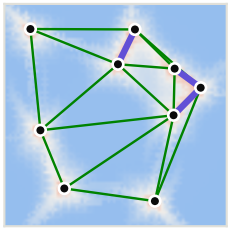
seed 10025 | precision 100% | gap 0.00%



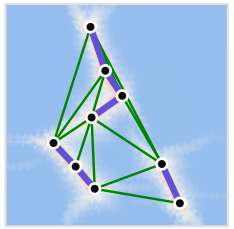
seed 10026 | precision 100% | gap 0.00%



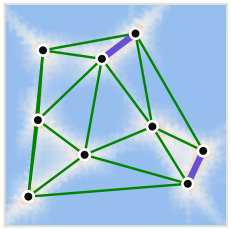
seed 10027 | precision 100% | gap 0.00%



seed 10028 | precision 100% | gap 0.00%



seed 10029 | precision 100% | gap 0.00%

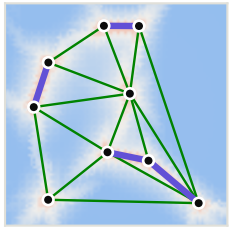


First 30 of 50 trials: cotton, 9 pillars

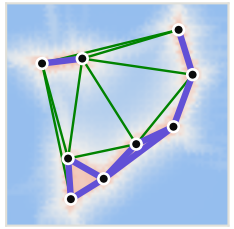
cloth edge in the MWT | cloth edge NOT in the MWT | MWT edge | cloth above threshold | cloth below threshold

50 layouts seen from above. Precision 97%, recall 40%, inside an MWT in 76% of trials, mean gap 0.479%.

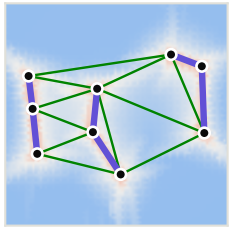
seed 10000 | precision 100% | gap 0.00%



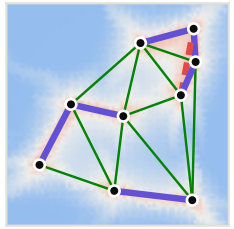
seed 10001 | precision 100% | gap 0.00%



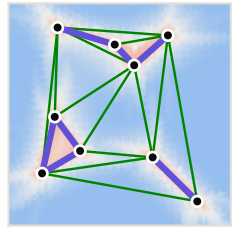
seed 10002 | precision 100% | gap 0.00%



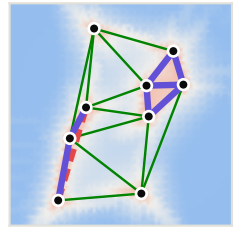
seed 10003 | precision 86% | gap 0.73%



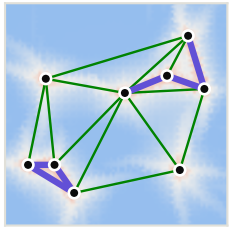
seed 10004 | precision 100% | gap 0.00%



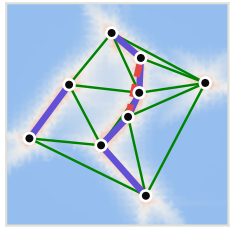
seed 10005 | precision 88% | gap 2.23%



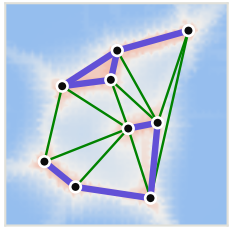
seed 10006 | precision 100% | gap 0.00%



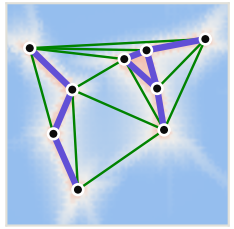
seed 10007 | precision 86% | gap 0.20%



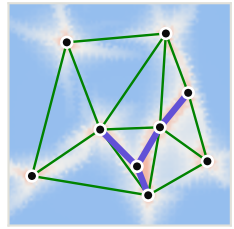
seed 10008 | precision 100% | gap 0.00%



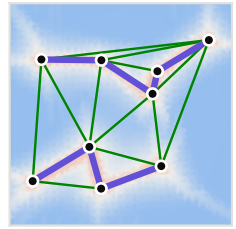
seed 10009 | precision 100% | gap 0.00%



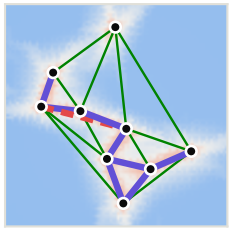
seed 10010 | precision 100% | gap 0.00%



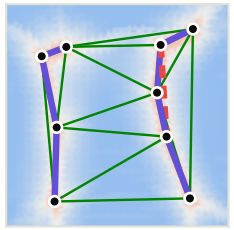
seed 10011 | precision 100% | gap 0.00%



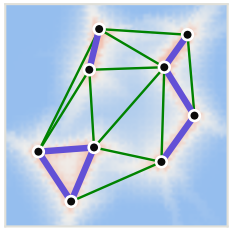
seed 10012 | precision 89% | gap 2.62%



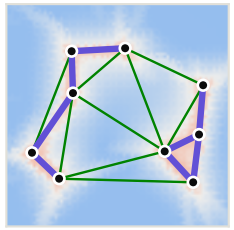
seed 10013 | precision 88% | gap 1.08%



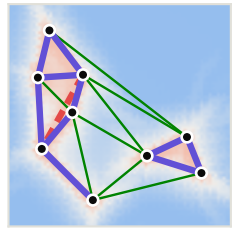
seed 10014 | precision 100% | gap 0.00%



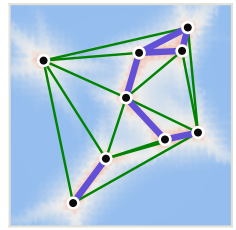
seed 10015 | precision 100% | gap 0.00%



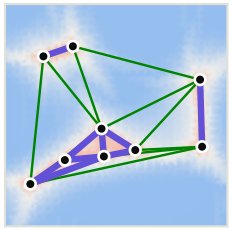
seed 10016 | precision 91% | gap 2.71%



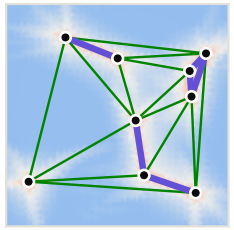
seed 10017 | precision 100% | gap 0.00%



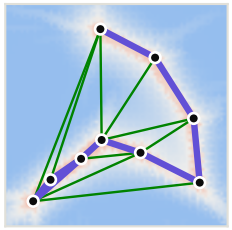
seed 10018 | precision 100% | gap 0.00%



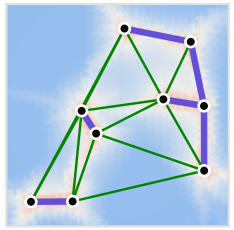
seed 10019 | precision 100% | gap 0.00%



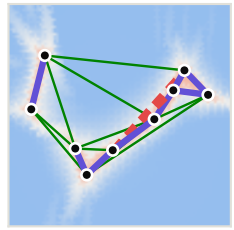
seed 10020 | precision 100% | gap 0.00%



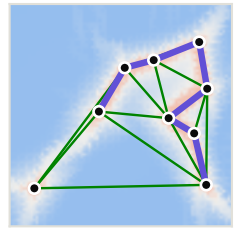
seed 10021 | precision 100% | gap 0.00%



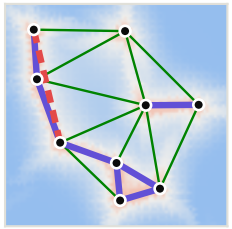
seed 10022 | precision 80% | gap 2.65%



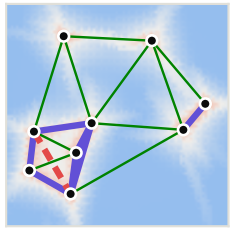
seed 10023 | precision 100% | gap 0.00%



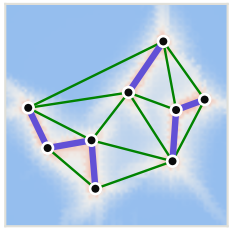
seed 10024 | precision 88% | gap 2.61%



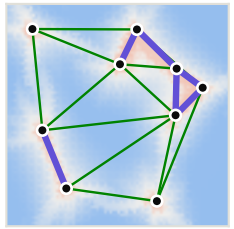
seed 10025 | precision 88% | gap 1.85%



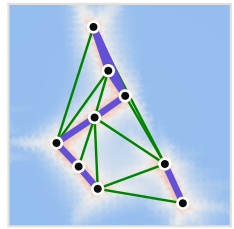
seed 10026 | precision 100% | gap 0.00%



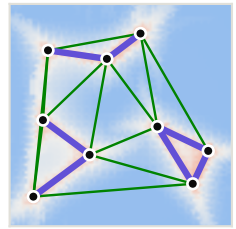
seed 10027 | precision 100% | gap 0.00%



seed 10028 | precision 100% | gap 0.00%



seed 10029 | precision 100% | gap 0.00%

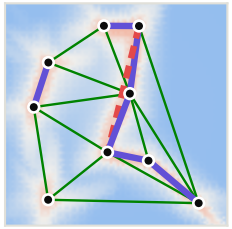


First 30 of 50 trials: canvas, 9 pillars

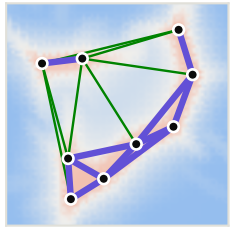
cloth edge in the MWT | cloth edge NOT in the MWT | MWT edge | cloth above threshold | cloth below threshold

50 layouts seen from above. Precision 92%, recall 47%, inside an MWT in 42% of trials, mean gap 2.194%.

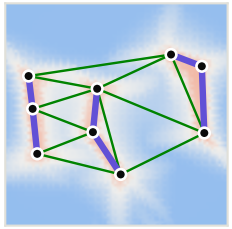
seed 10000 | precision 86% | gap 5.09%



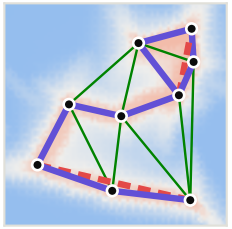
seed 10001 | precision 100% | gap 0.00%



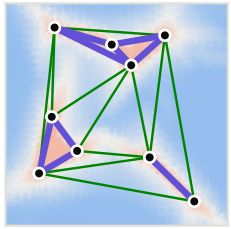
seed 10002 | precision 100% | gap 0.00%



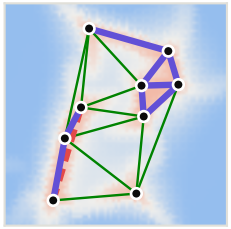
seed 10003 | precision 82% | gap 7.12%



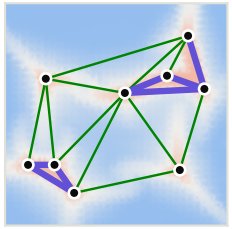
seed 10004 | precision 100% | gap 0.00%



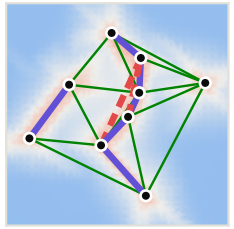
seed 10005 | precision 89% | gap 2.23%



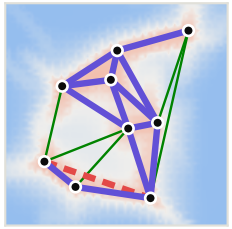
seed 10006 | precision 100% | gap 0.00%



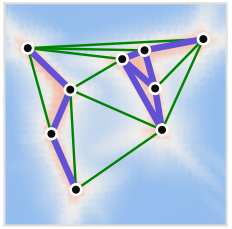
seed 10007 | precision 75% | gap 2.33%



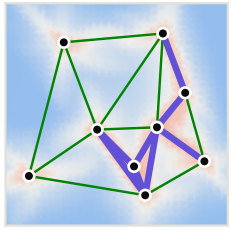
seed 10008 | precision 93% | gap 2.57%



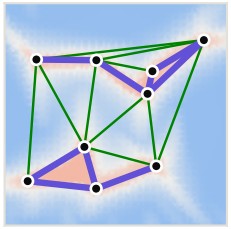
seed 10009 | precision 100% | gap 0.00%



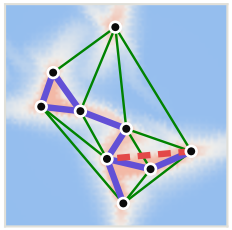
seed 10010 | precision 100% | gap 0.00%



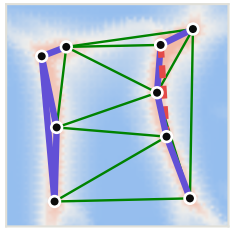
seed 10011 | precision 100% | gap 0.00%



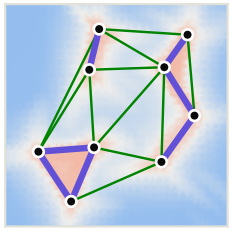
seed 10012 | precision 89% | gap 2.95%



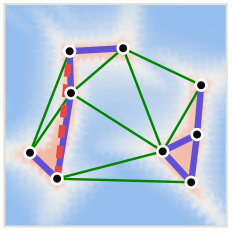
seed 10013 | precision 89% | gap 1.08%



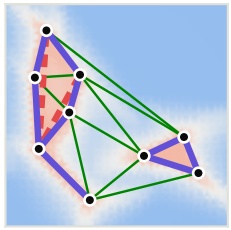
seed 10014 | precision 100% | gap 0.00%



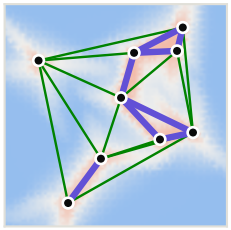
seed 10015 | precision 89% | gap 4.38%



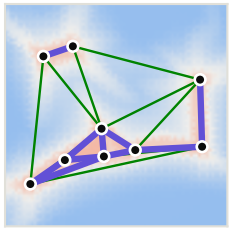
seed 10016 | precision 82% | gap 8.25%



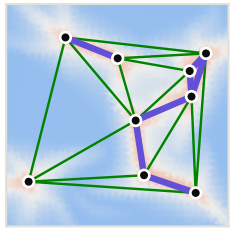
seed 10017 | precision 100% | gap 0.00%



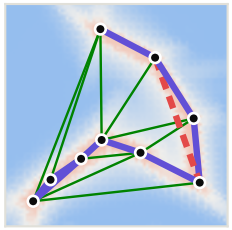
seed 10018 | precision 100% | gap 0.00%



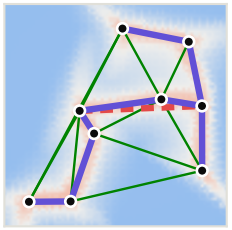
seed 10019 | precision 100% | gap 0.00%



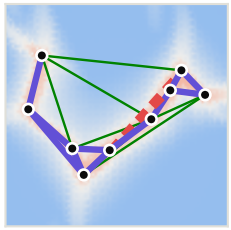
seed 10020 | precision 90% | gap 4.50%



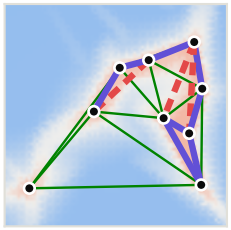
seed 10021 | precision 89% | gap 5.08%



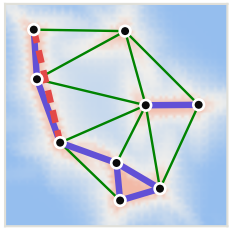
seed 10022 | precision 85% | gap 2.65%



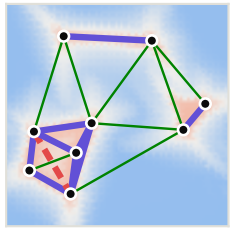
seed 10023 | precision 73% | gap 5.13%



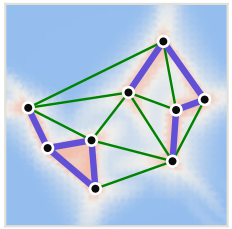
seed 10024 | precision 88% | gap 2.61%



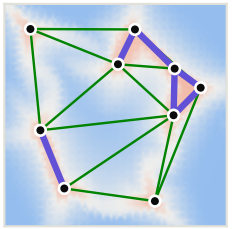
seed 10025 | precision 90% | gap 1.85%



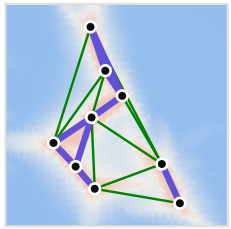
seed 10026 | precision 100% | gap 0.00%



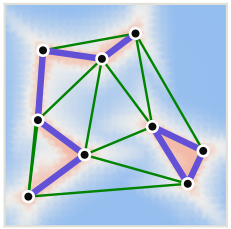
seed 10027 | precision 100% | gap 0.00%



seed 10028 | precision 100% | gap 0.00%



seed 10029 | precision 100% | gap 0.00%

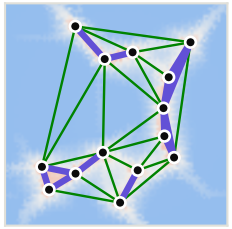


First 30 of 50 trials: silk, 14 pillars

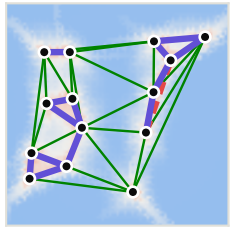
cloth edge in Greedy | cloth edge NOT in Greedy | Greedy edge | cloth above threshold | cloth below threshold

50 layouts seen from above, compared with Greedy. The cloth's edges + exact completion was shorter in 5, the same in 23, longer in 22; mean +0.513%.

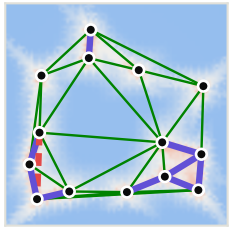
seed 15000 | cloth same as Greedy



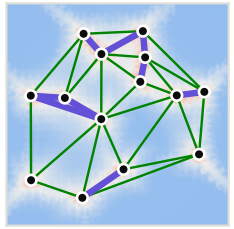
seed 15001 | cloth +0.04% vs Greedy



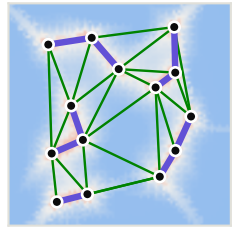
seed 15002 | cloth +0.87% vs Greedy



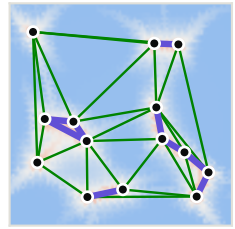
seed 15003 | cloth same as Greedy



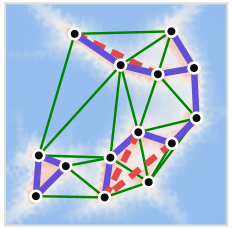
seed 15004 | cloth same as Greedy



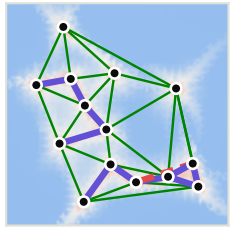
seed 15005 | cloth same as Greedy



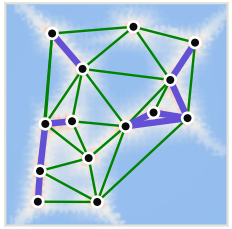
seed 15006 | cloth +5.31% vs Greedy



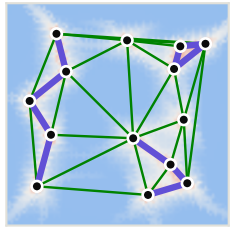
seed 15007 | cloth -0.64% vs Greedy



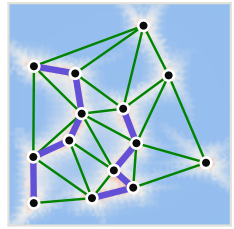
seed 15008 | cloth same as Greedy



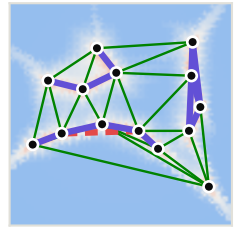
seed 15009 | cloth same as Greedy



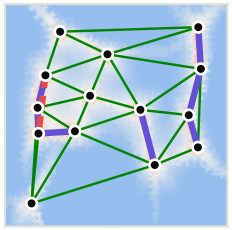
seed 15010 | cloth -0.14% vs Greedy



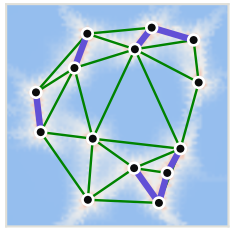
seed 15011 | cloth +2.21% vs Greedy



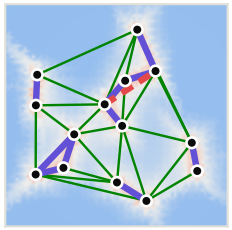
seed 15012 | cloth +1.22% vs Greedy



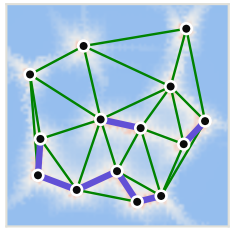
seed 15013 | cloth same as Greedy



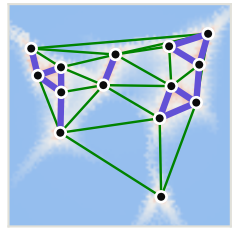
seed 15014 | cloth +0.81% vs Greedy



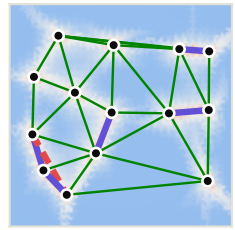
seed 15015 | cloth same as Greedy



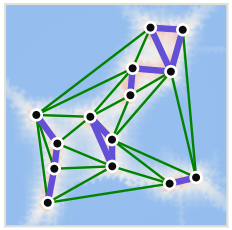
seed 15016 | cloth same as Greedy



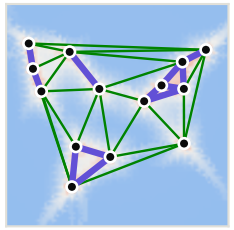
seed 15017 | cloth +0.69% vs Greedy



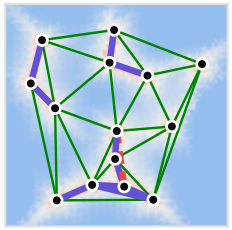
seed 15018 | cloth same as Greedy



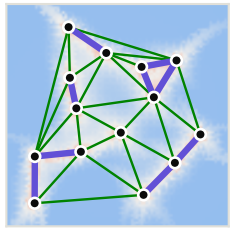
seed 15019 | cloth same as Greedy



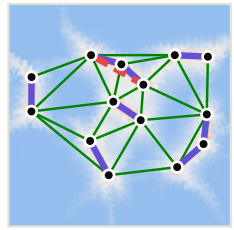
seed 15020 | cloth +0.56% vs Greedy



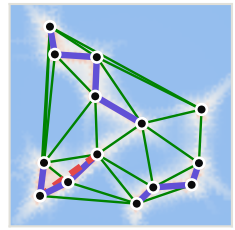
seed 15021 | cloth same as Greedy



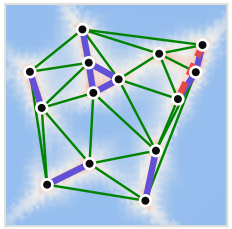
seed 15022 | cloth +1.36% vs Greedy



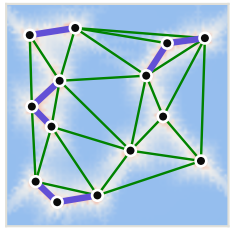
seed 15023 | cloth +1.82% vs Greedy



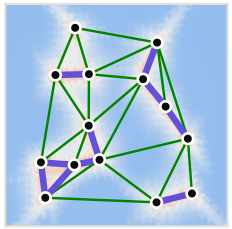
seed 15024 | cloth +0.88% vs Greedy



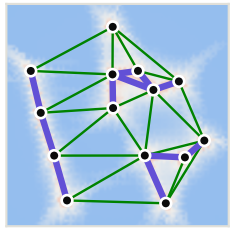
seed 15025 | cloth same as Greedy



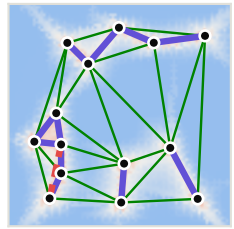
seed 15026 | cloth -0.11% vs Greedy



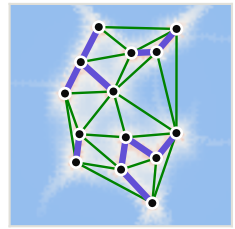
seed 15027 | cloth same as Greedy



seed 15028 | cloth +0.64% vs Greedy



seed 15029 | cloth same as Greedy

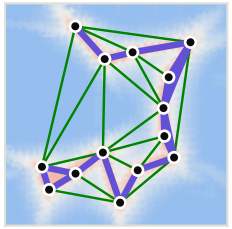


First 30 of 50 trials: cotton, 14 pillars

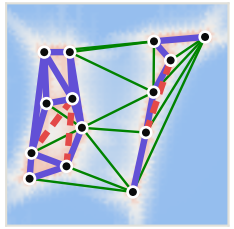
cloth edge in Greedy | cloth edge NOT in Greedy | Greedy edge | cloth above threshold | cloth below threshold

50 layouts seen from above, compared with Greedy. The cloth's edges + exact completion was shorter in 0, the same in 13, longer in 37; mean +1.794%.

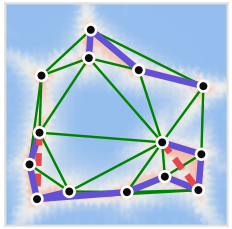
seed 15000 | cloth same as Greedy



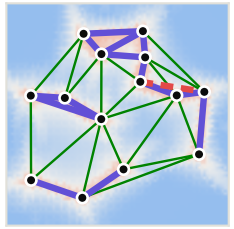
seed 15001 | cloth +1.81% vs Greedy



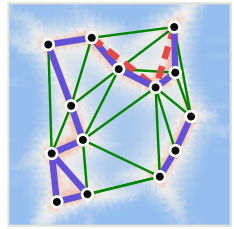
seed 15002 | cloth +1.67% vs Greedy



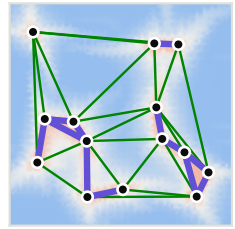
seed 15003 | cloth +0.79% vs Greedy



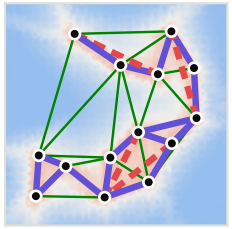
seed 15004 | cloth +0.96% vs Greedy



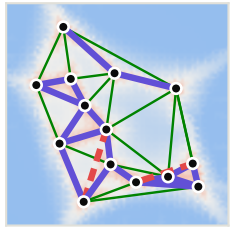
seed 15005 | cloth same as Greedy



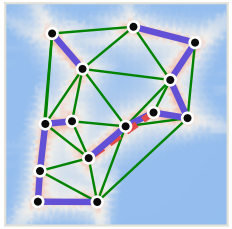
seed 15006 | cloth +8.33% vs Greedy



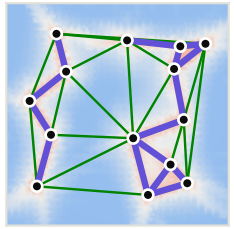
seed 15007 | cloth +0.45% vs Greedy



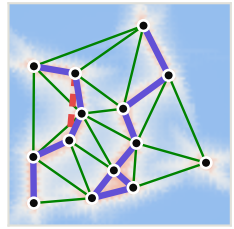
seed 15008 | cloth +2.14% vs Greedy



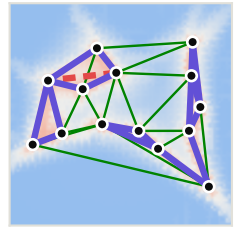
seed 15009 | cloth same as Greedy



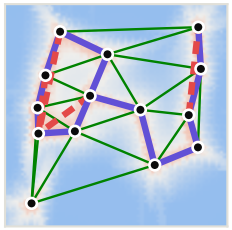
seed 15010 | cloth +0.80% vs Greedy



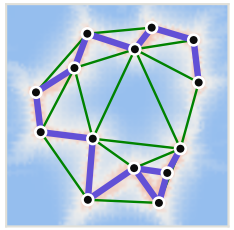
seed 15011 | cloth +1.31% vs Greedy



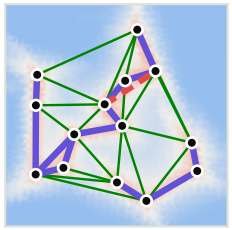
seed 15012 | cloth +5.46% vs Greedy



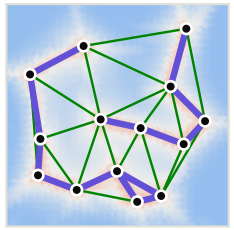
seed 15013 | cloth same as Greedy



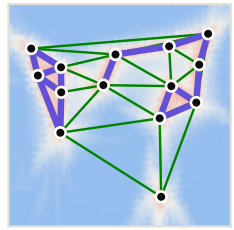
seed 15014 | cloth +0.81% vs Greedy



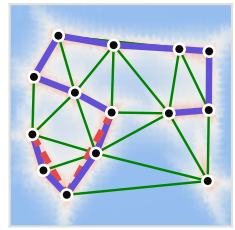
seed 15015 | cloth same as Greedy



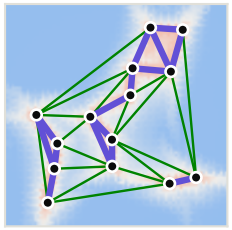
seed 15016 | cloth same as Greedy



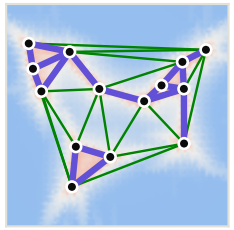
seed 15017 | cloth +2.87% vs Greedy



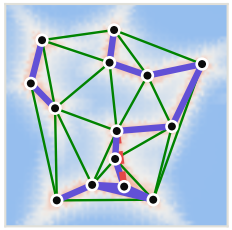
seed 15018 | cloth same as Greedy



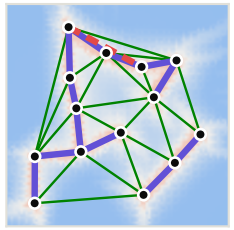
seed 15019 | cloth same as Greedy



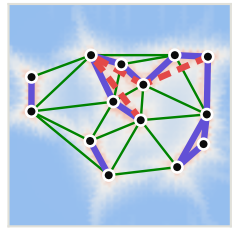
seed 15020 | cloth +0.56% vs Greedy



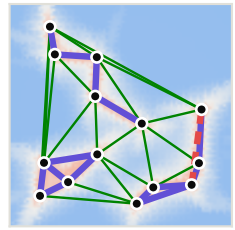
seed 15021 | cloth +0.64% vs Greedy



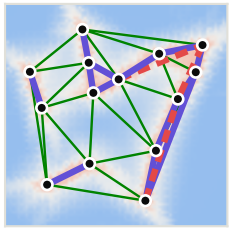
seed 15022 | cloth +4.47% vs Greedy



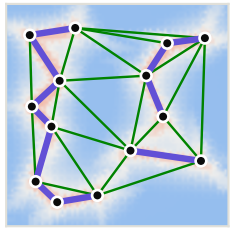
seed 15023 | cloth +1.54% vs Greedy



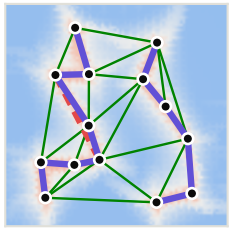
seed 15024 | cloth +3.76% vs Greedy



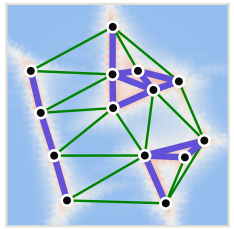
seed 15025 | cloth same as Greedy



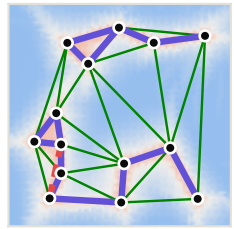
seed 15026 | cloth +2.72% vs Greedy



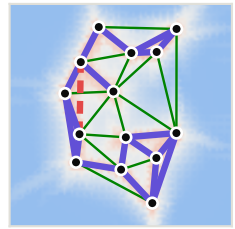
seed 15027 | cloth same as Greedy



seed 15028 | cloth +0.64% vs Greedy



seed 15029 | cloth +1.44% vs Greedy

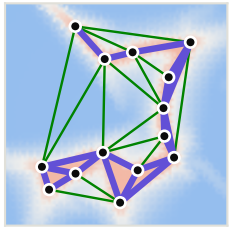


First 30 of 50 trials: canvas, 14 pillars

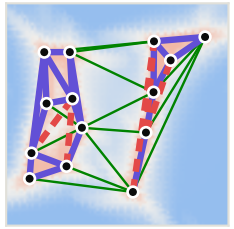
cloth edge in Greedy | cloth edge NOT in Greedy | Greedy edge | cloth above threshold | cloth below threshold

50 layouts seen from above, compared with Greedy. The cloth's edges + exact completion was shorter in 0, the same in 5, longer in 45; mean +3.120%.

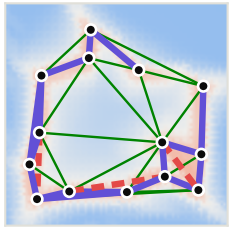
seed 15000 | cloth same as Greedy



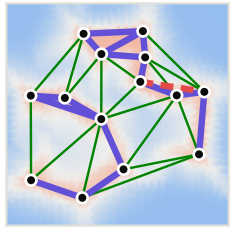
seed 15001 | cloth +7.59% vs Greedy



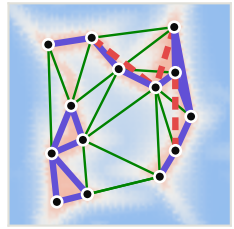
seed 15002 | cloth +3.34% vs Greedy



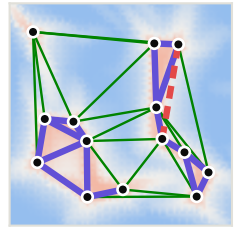
seed 15003 | cloth +0.79% vs Greedy



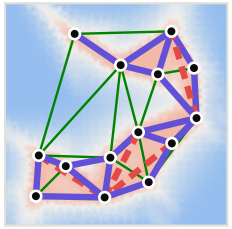
seed 15004 | cloth +2.68% vs Greedy



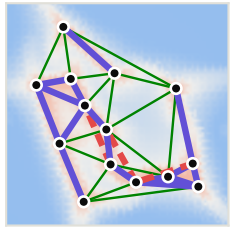
seed 15005 | cloth +3.01% vs Greedy



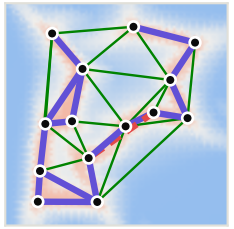
seed 15006 | cloth +8.56% vs Greedy



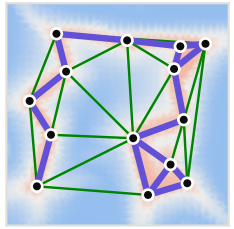
seed 15007 | cloth +0.17% vs Greedy



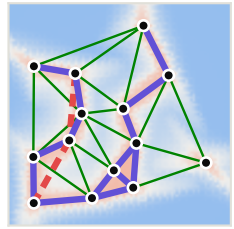
seed 15008 | cloth +2.14% vs Greedy



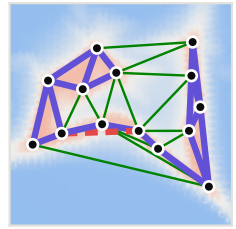
seed 15009 | cloth same as Greedy



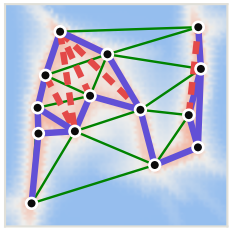
seed 15010 | cloth +0.82% vs Greedy



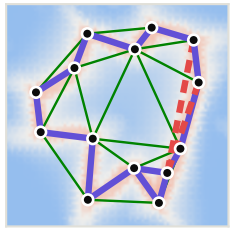
seed 15011 | cloth +2.21% vs Greedy



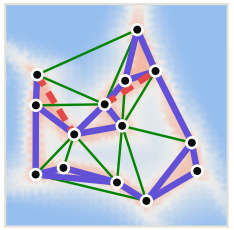
seed 15012 | cloth +7.93% vs Greedy



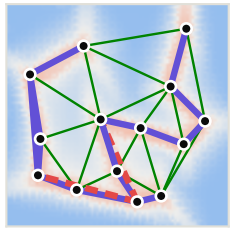
seed 15013 | cloth +6.93% vs Greedy



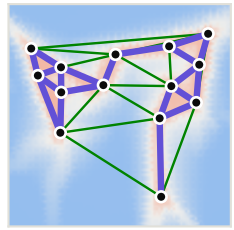
seed 15014 | cloth +0.88% vs Greedy



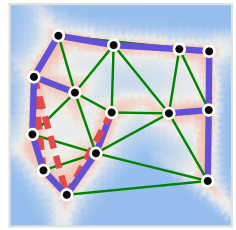
seed 15015 | cloth +7.74% vs Greedy



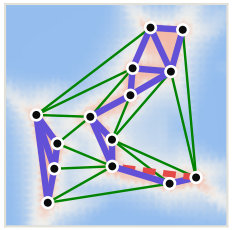
seed 15016 | cloth same as Greedy



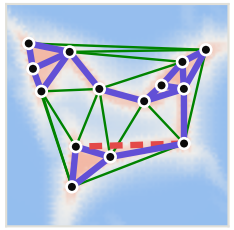
seed 15017 | cloth +7.22% vs Greedy



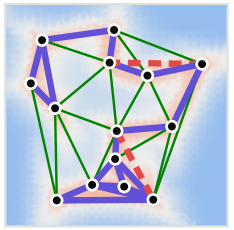
seed 15018 | cloth +0.41% vs Greedy



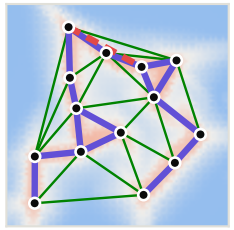
seed 15019 | cloth +2.63% vs Greedy



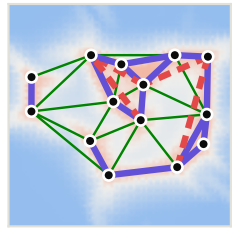
seed 15020 | cloth +2.33% vs Greedy



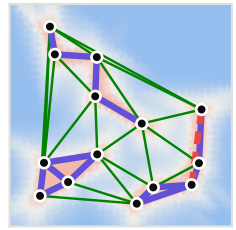
seed 15021 | cloth +0.64% vs Greedy



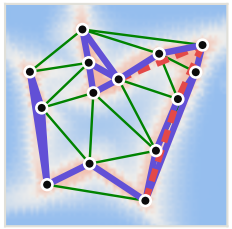
seed 15022 | cloth +8.64% vs Greedy



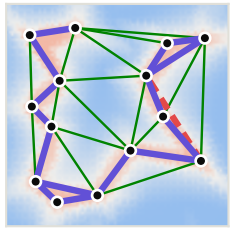
seed 15023 | cloth +1.54% vs Greedy



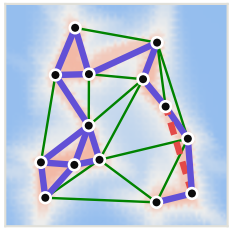
seed 15024 | cloth +3.76% vs Greedy



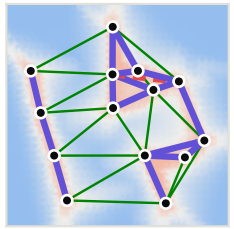
seed 15025 | cloth +0.60% vs Greedy



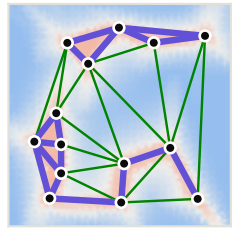
seed 15026 | cloth +1.26% vs Greedy



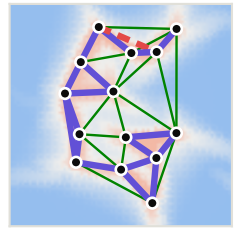
seed 15027 | cloth +2.22% vs Greedy



seed 15028 | cloth same as Greedy

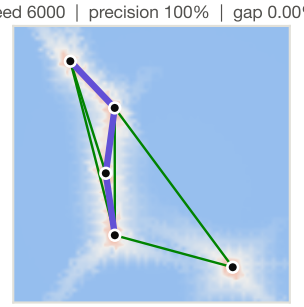
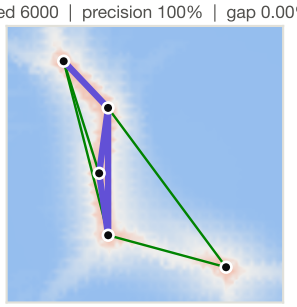
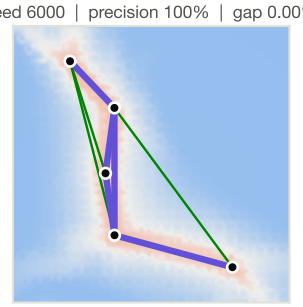
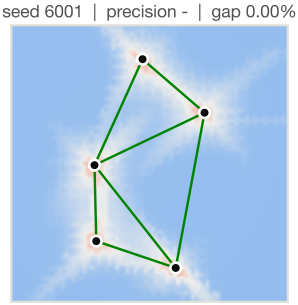
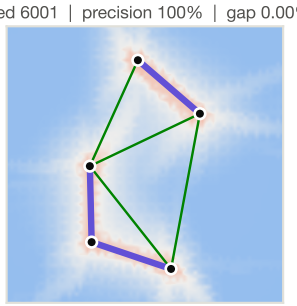
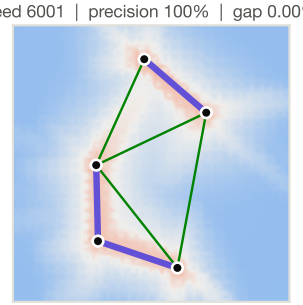
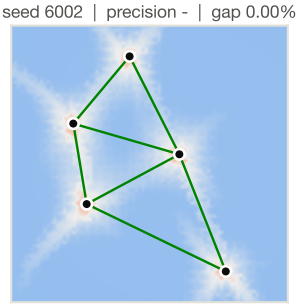
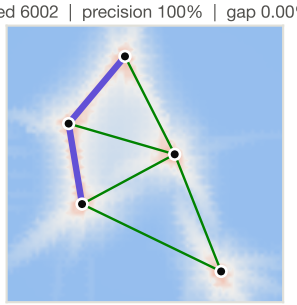
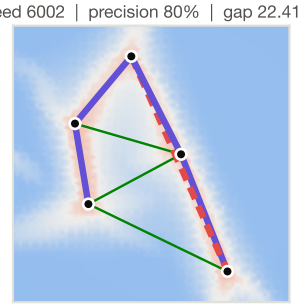
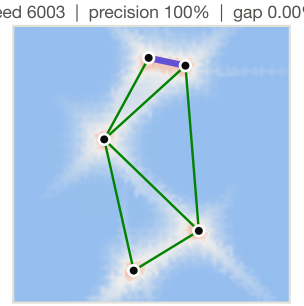
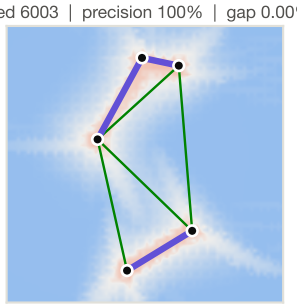
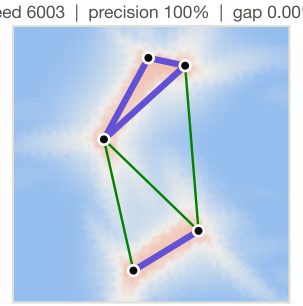
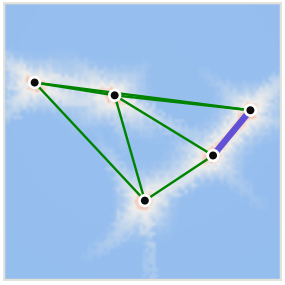
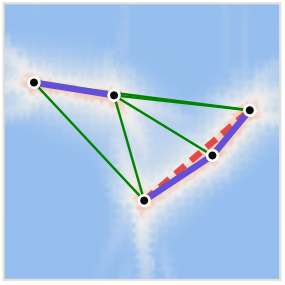
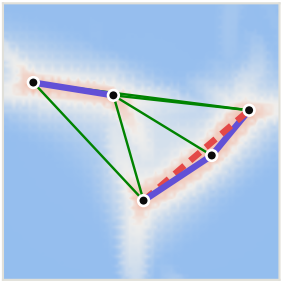
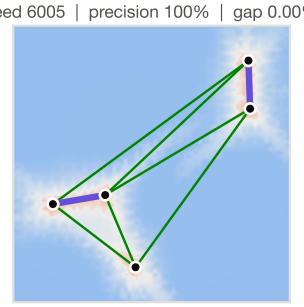
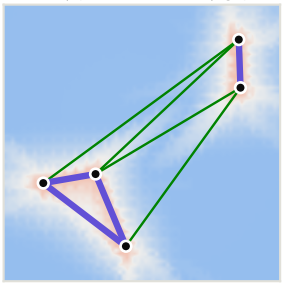
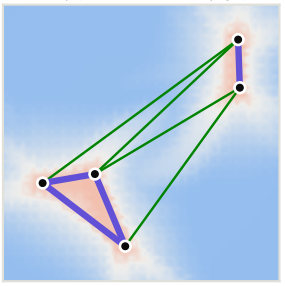


seed 15029 | cloth +0.72% vs Greedy



Same layout, different cloth: 5 pillars

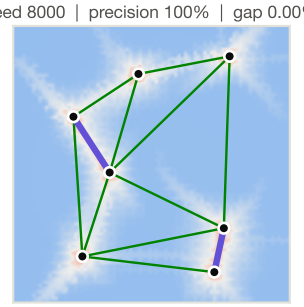
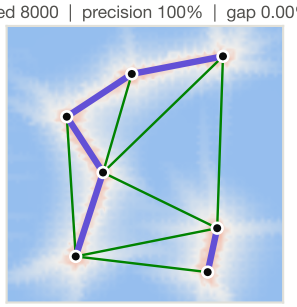
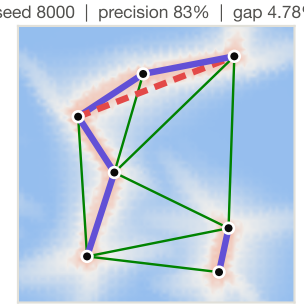
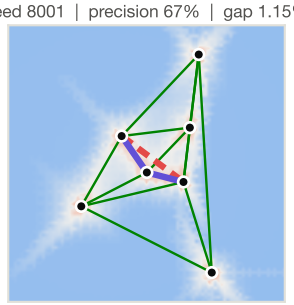
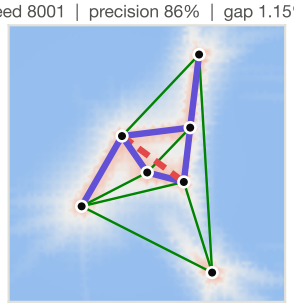
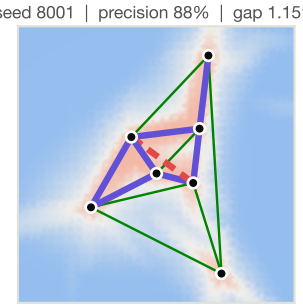
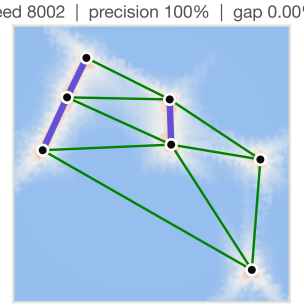
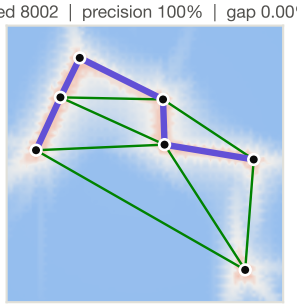
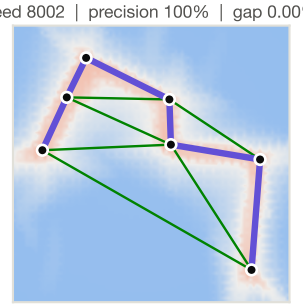
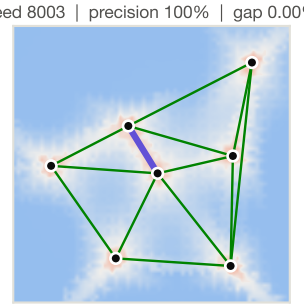
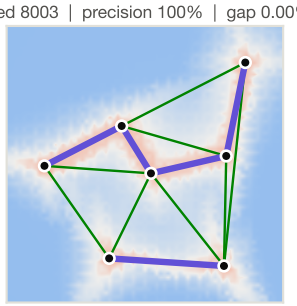
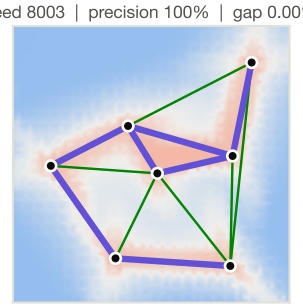
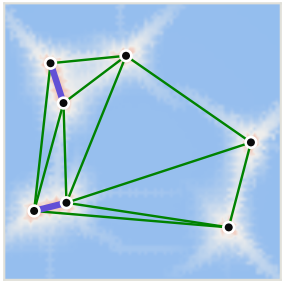
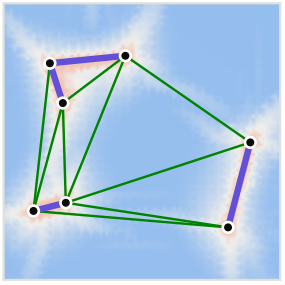
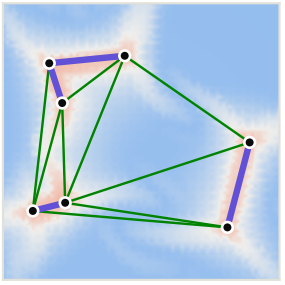
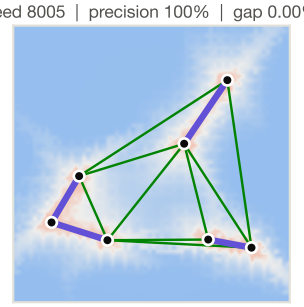
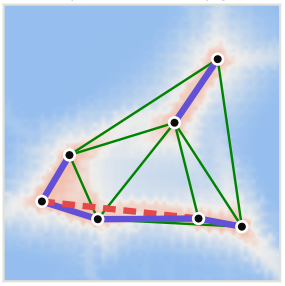
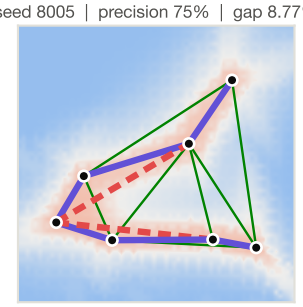
The first 6 layouts; each row is one layout dropped under every cloth type.

silk	cotton	canvas
seed 6000 | precision 100% | gap 0.00%	seed 6000 | precision 100% | gap 0.00%	seed 6000 | precision 100% | gap 0.00%
		
seed 6001 | precision - | gap 0.00%	seed 6001 | precision 100% | gap 0.00%	seed 6001 | precision 100% | gap 0.00%
		
seed 6002 | precision - | gap 0.00%	seed 6002 | precision 100% | gap 0.00%	seed 6002 | precision 80% | gap 22.41%
		
seed 6003 | precision 100% | gap 0.00%	seed 6003 | precision 100% | gap 0.00%	seed 6003 | precision 100% | gap 0.00%
		
seed 6004 | precision 100% | gap 0.00%	seed 6004 | precision 75% | gap 2.46%	seed 6004 | precision 75% | gap 2.46%
		
seed 6005 | precision 100% | gap 0.00%	seed 6005 | precision 100% | gap 0.00%	seed 6005 | precision 100% | gap 0.00%
		

cloth edge in the MWT — cloth edge NOT in the MWT — MWT edge

Same layout, different cloth: 7 pillars

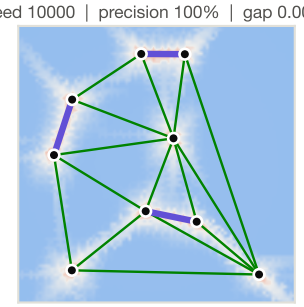
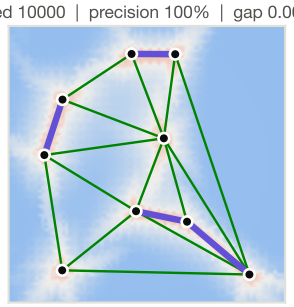
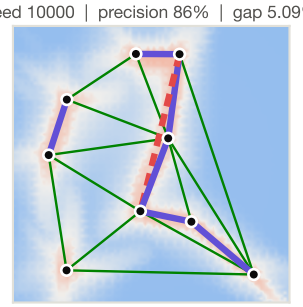
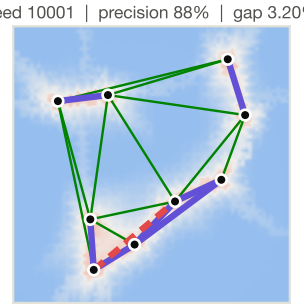
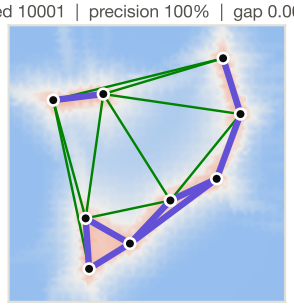
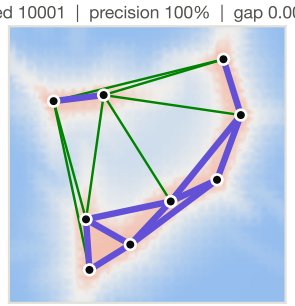
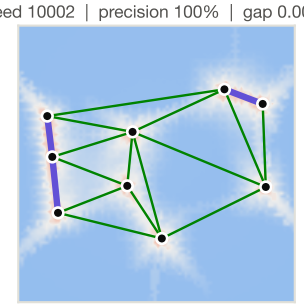
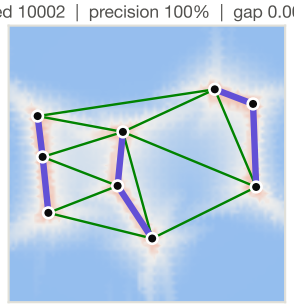
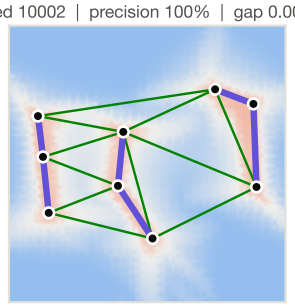
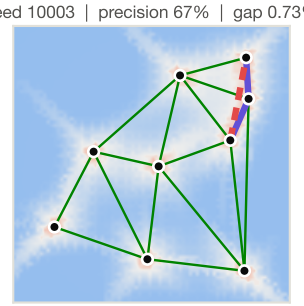
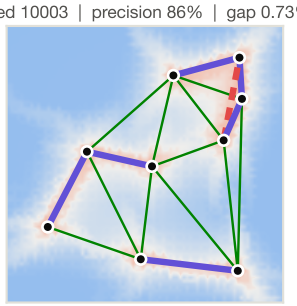
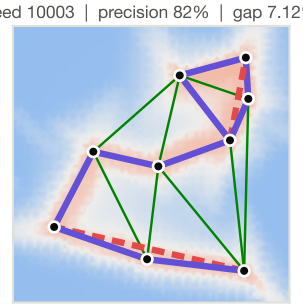
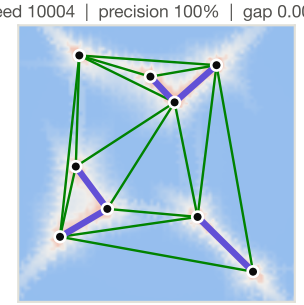
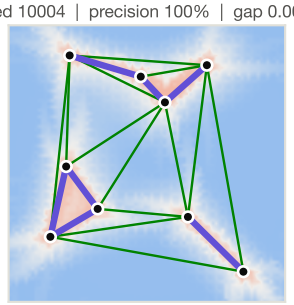
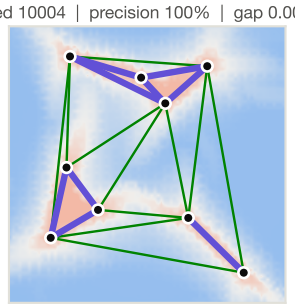
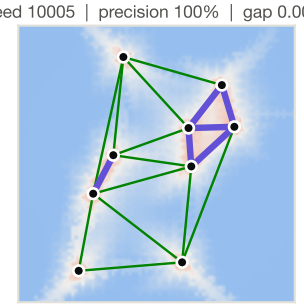
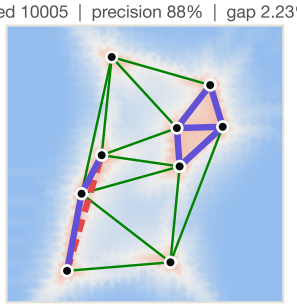
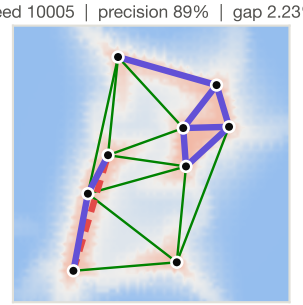
The first 6 layouts; each row is one layout dropped under every cloth type.

silk	cotton	canvas
seed 8000 | precision 100% | gap 0.00%	seed 8000 | precision 100% | gap 0.00%	seed 8000 | precision 83% | gap 4.78%
		
seed 8001 | precision 67% | gap 1.15%	seed 8001 | precision 86% | gap 1.15%	seed 8001 | precision 88% | gap 1.15%
		
seed 8002 | precision 100% | gap 0.00%	seed 8002 | precision 100% | gap 0.00%	seed 8002 | precision 100% | gap 0.00%
		
seed 8003 | precision 100% | gap 0.00%	seed 8003 | precision 100% | gap 0.00%	seed 8003 | precision 100% | gap 0.00%
		
seed 8004 | precision 100% | gap 0.00%	seed 8004 | precision 100% | gap 0.00%	seed 8004 | precision 100% | gap 0.00%
		
seed 8005 | precision 100% | gap 0.00%	seed 8005 | precision 83% | gap 8.01%	seed 8005 | precision 75% | gap 8.77%
		

cloth edge in the MWT — cloth edge NOT in the MWT — MWT edge

Same layout, different cloth: 9 pillars

The first 6 layouts; each row is one layout dropped under every cloth type.

silk	cotton	canvas
seed 10000 | precision 100% | gap 0.00%	seed 10000 | precision 100% | gap 0.00%	seed 10000 | precision 86% | gap 5.09%
		
seed 10001 | precision 88% | gap 3.20%	seed 10001 | precision 100% | gap 0.00%	seed 10001 | precision 100% | gap 0.00%
		
seed 10002 | precision 100% | gap 0.00%	seed 10002 | precision 100% | gap 0.00%	seed 10002 | precision 100% | gap 0.00%
		
seed 10003 | precision 67% | gap 0.73%	seed 10003 | precision 86% | gap 0.73%	seed 10003 | precision 82% | gap 7.12%
		
seed 10004 | precision 100% | gap 0.00%	seed 10004 | precision 100% | gap 0.00%	seed 10004 | precision 100% | gap 0.00%
		
seed 10005 | precision 100% | gap 0.00%	seed 10005 | precision 88% | gap 2.23%	seed 10005 | precision 89% | gap 2.23%
		

cloth edge in the MWT — cloth edge NOT in the MWT — MWT edge